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Tsai et al.

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(54) **IMAGING OPTICAL LENS SYSTEM,
IMAGING DEVICE AND MOBILE TERMINAL**

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U.S.C. 154(b) by 0 days.

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(57) **ABSTRACT**

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G02B 13/18 (2006.01)

H04N 5/225 (2006.01)

(52) **U.S. Cl.**

CPC **G02B 13/18** (2013.01); **G02B 9/60** (2013.01);

H04N 5/2254 (2013.01)

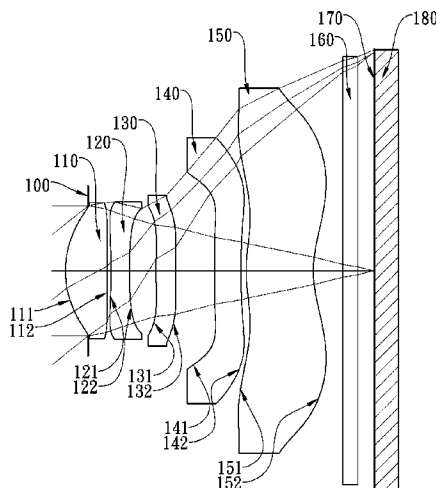
(58) **Field of Classification Search**

CPC H04N 5/2254; G02B 9/60; G02B 13/18

See application file for complete search history.

An imaging optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The fourth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The fifth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The imaging optical lens system has a total of five lens elements with refractive power.

34 Claims, 26 Drawing Sheets



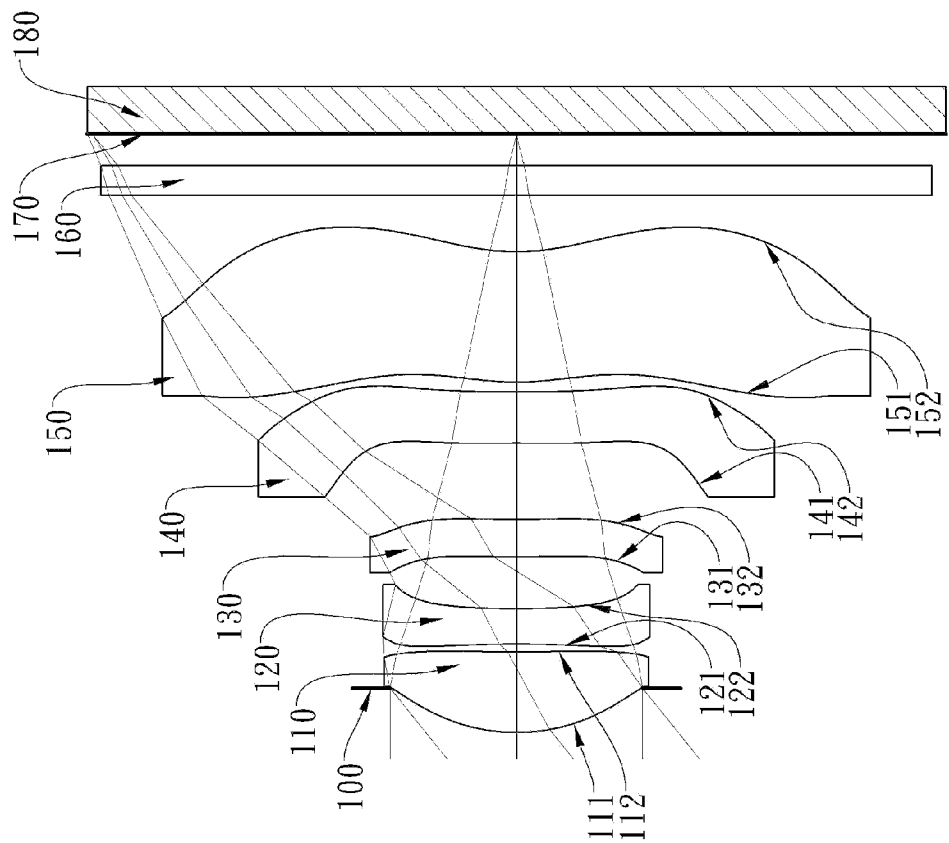


Fig. 1A

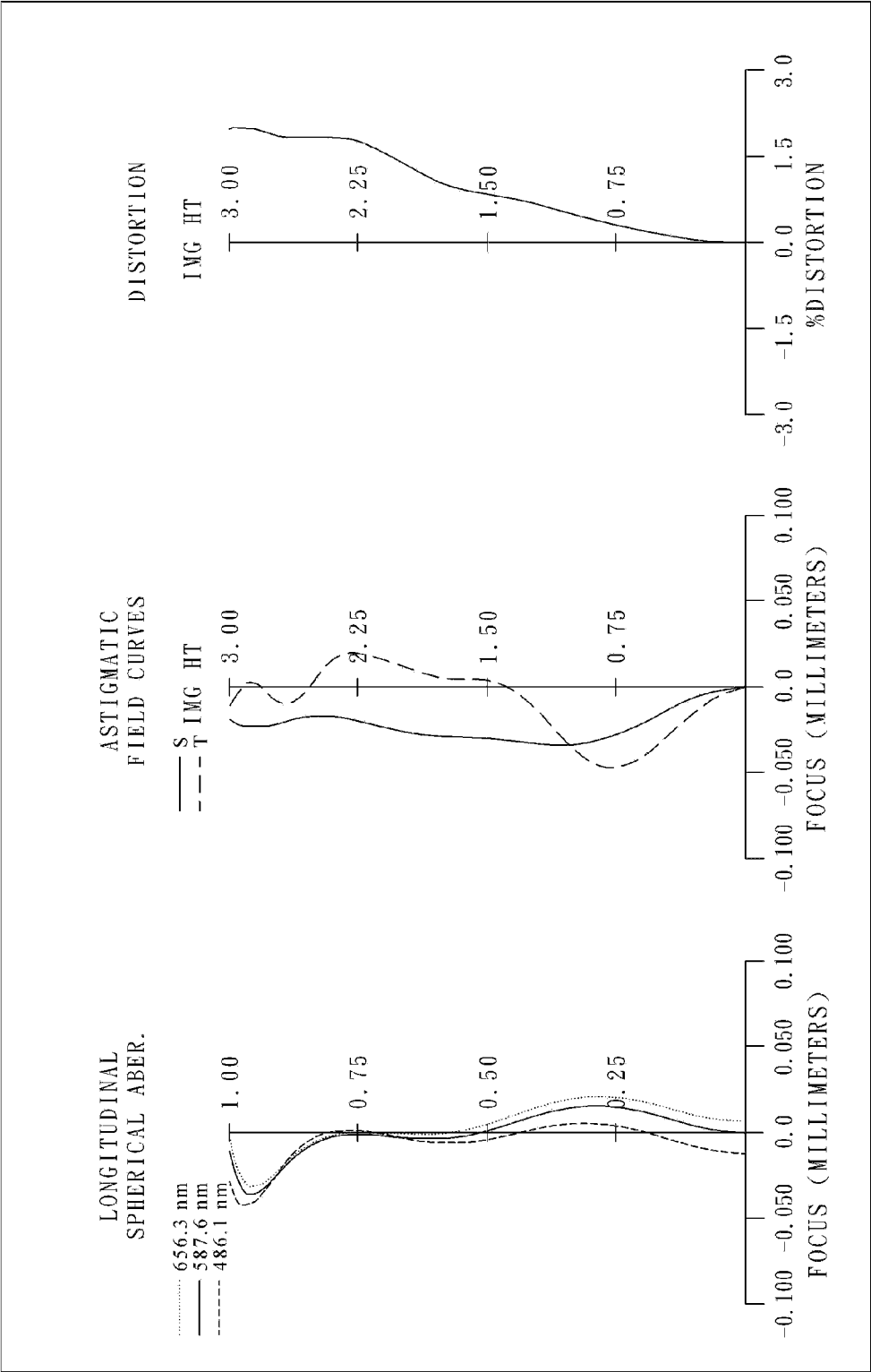


Fig. 1B

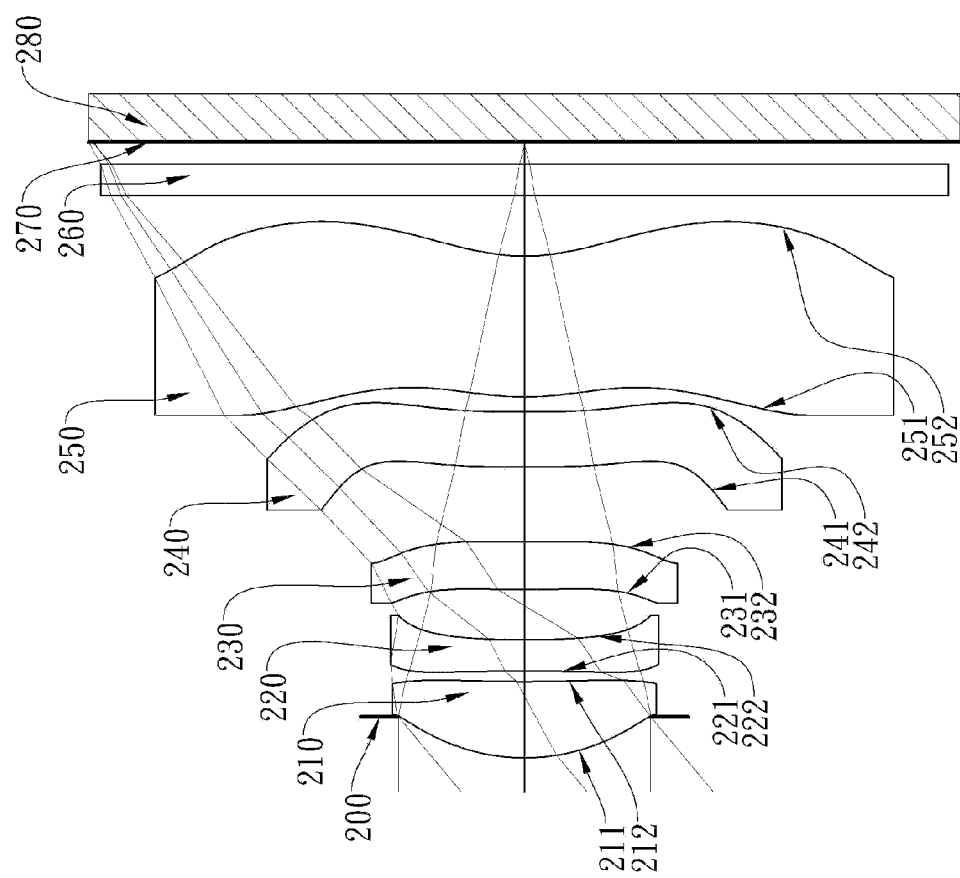


Fig. 2A

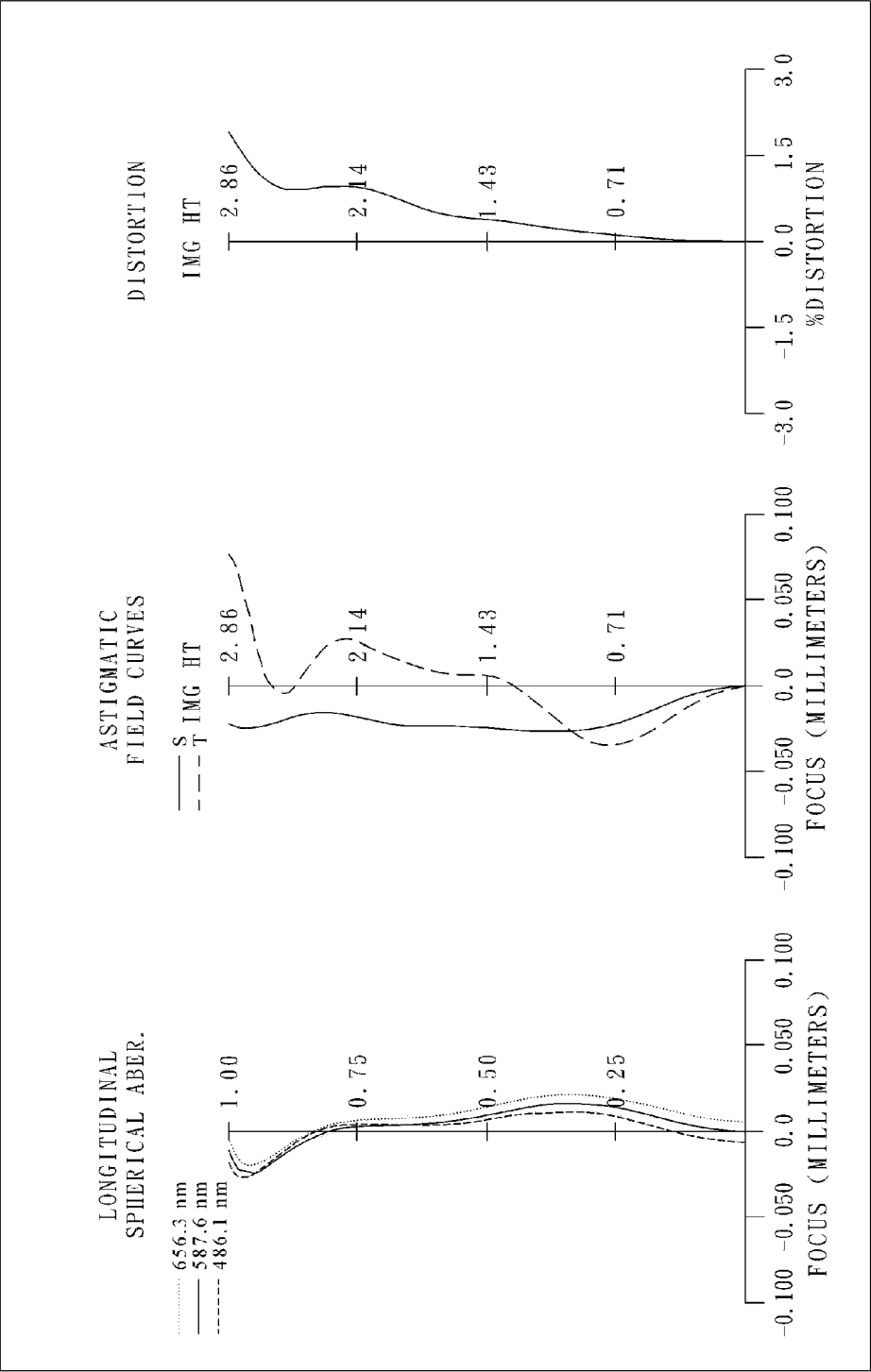


Fig. 2B

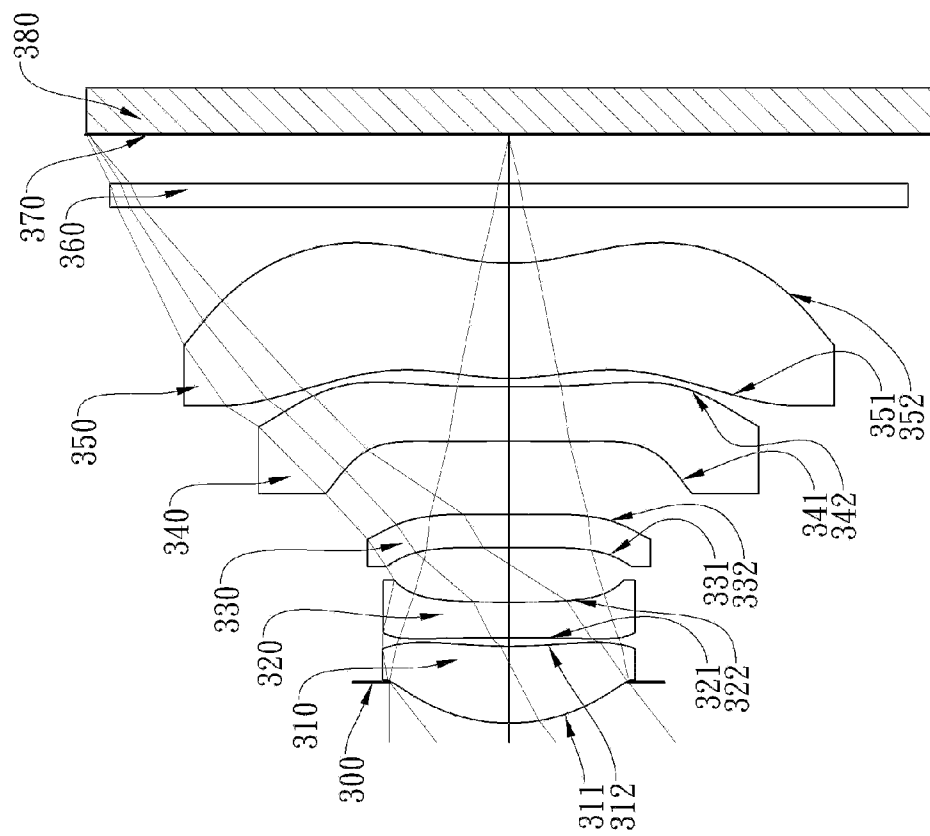


Fig. 3A

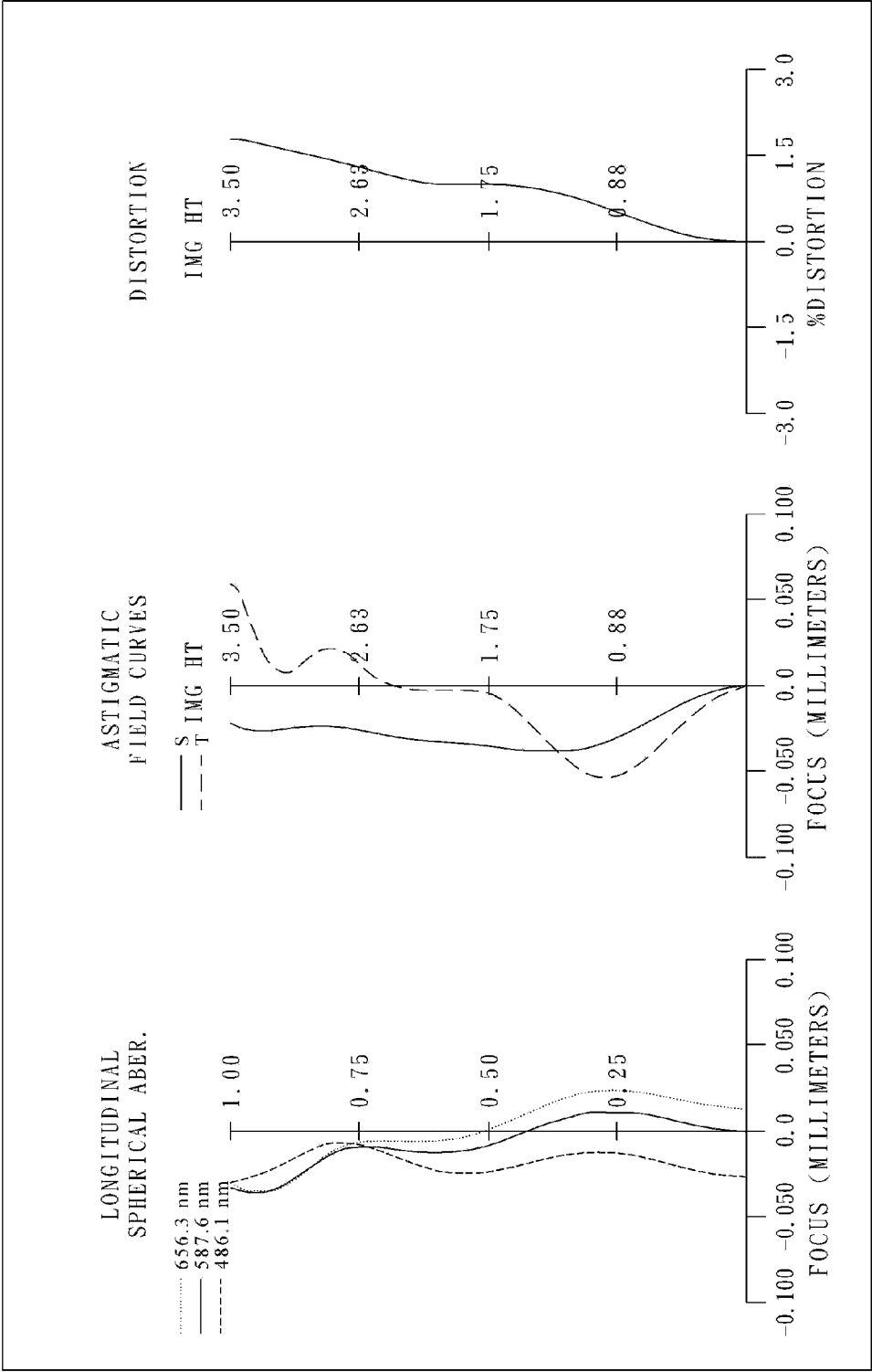


Fig. 3B

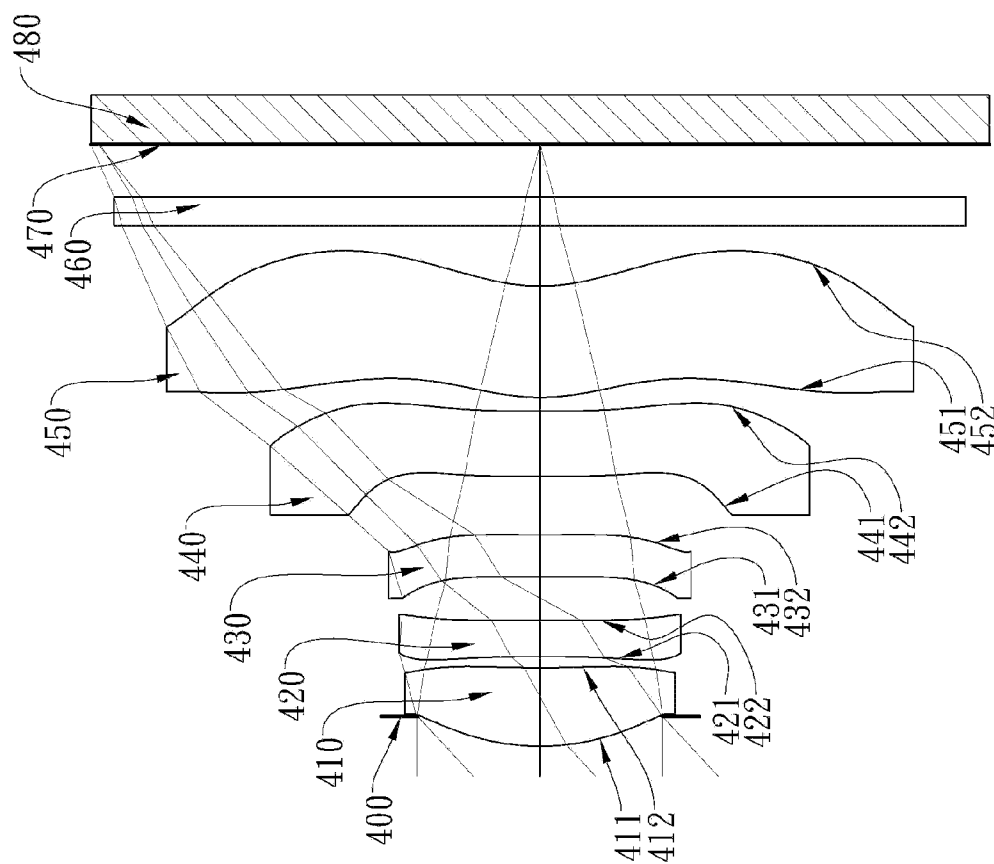


Fig. 4A

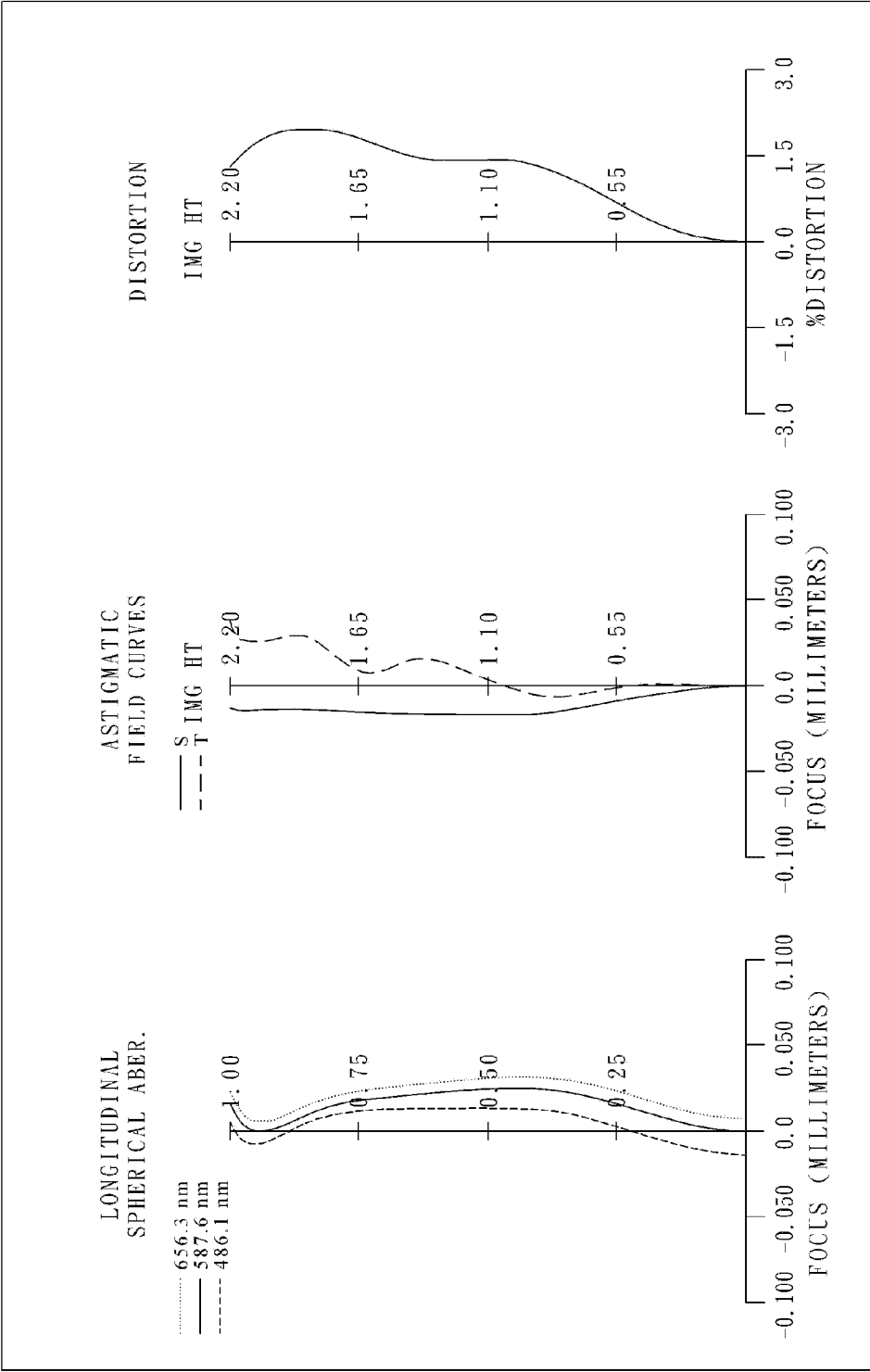


Fig. 4B

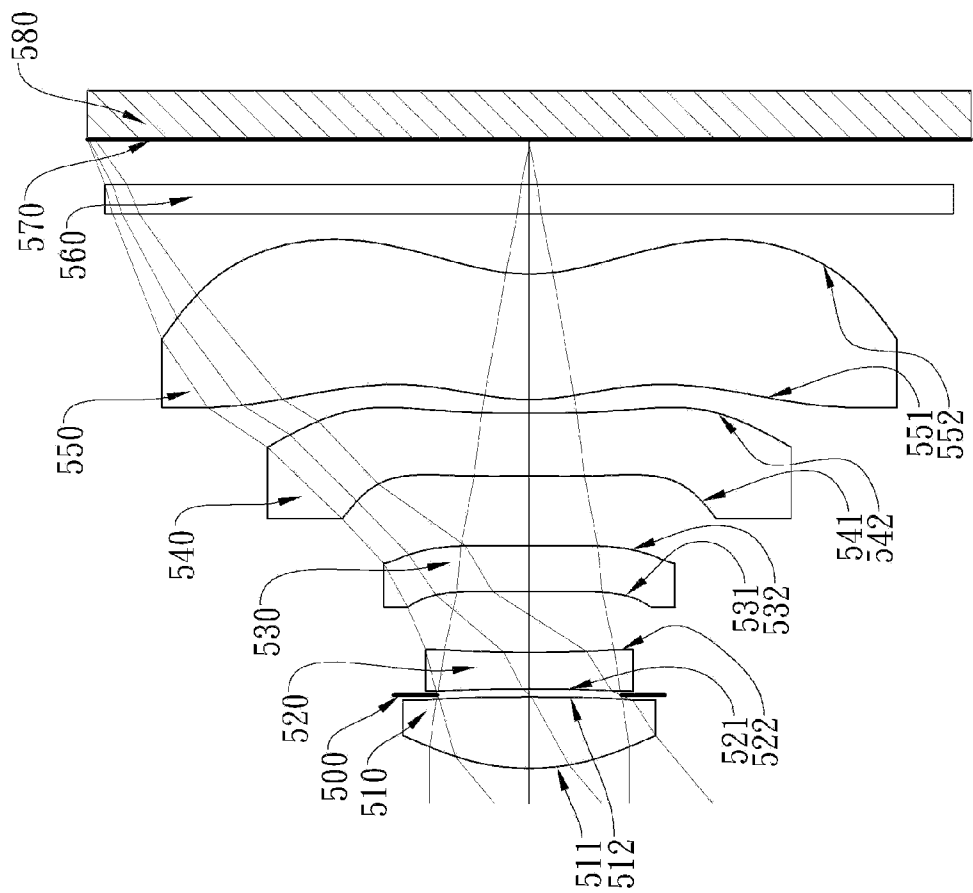


Fig. 5A

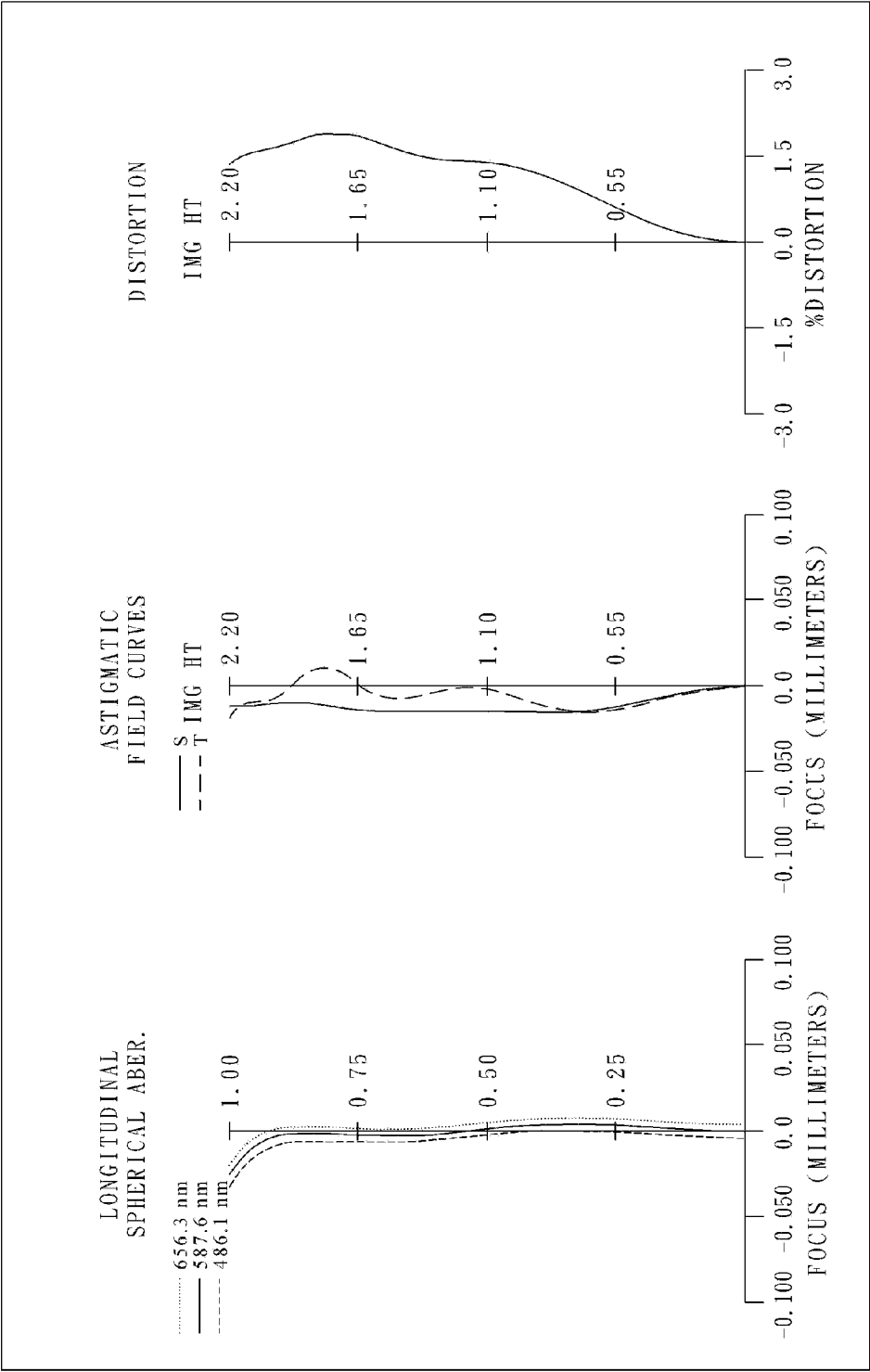


Fig. 5B

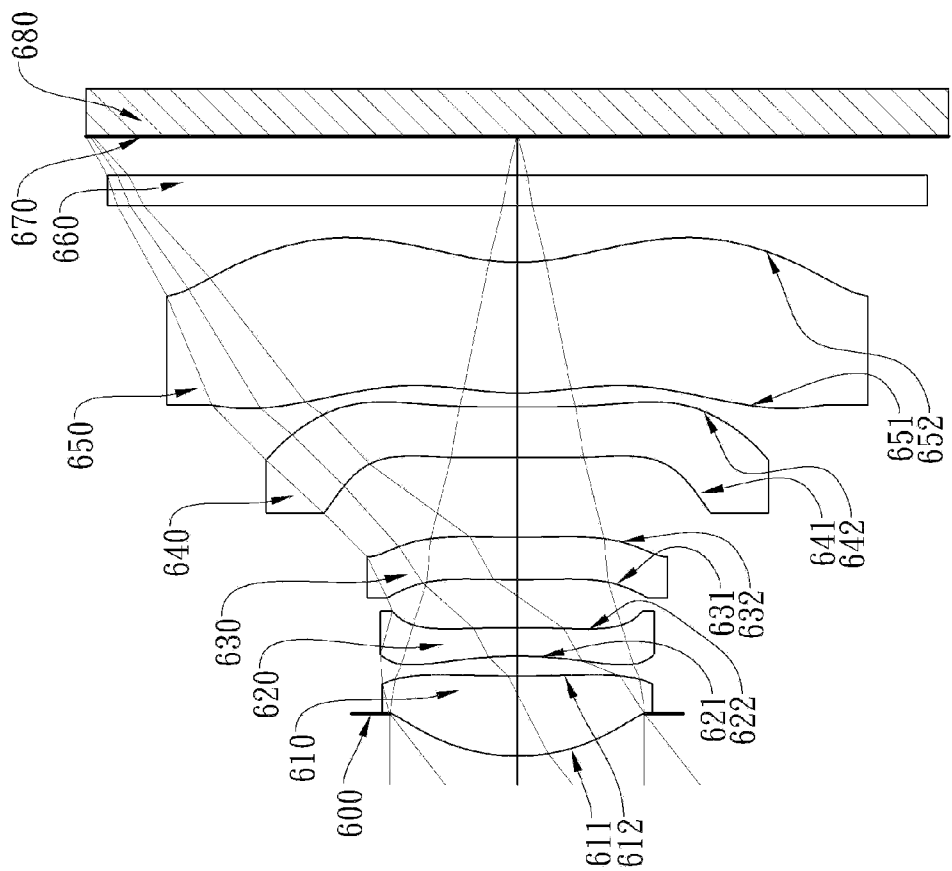


Fig. 6A

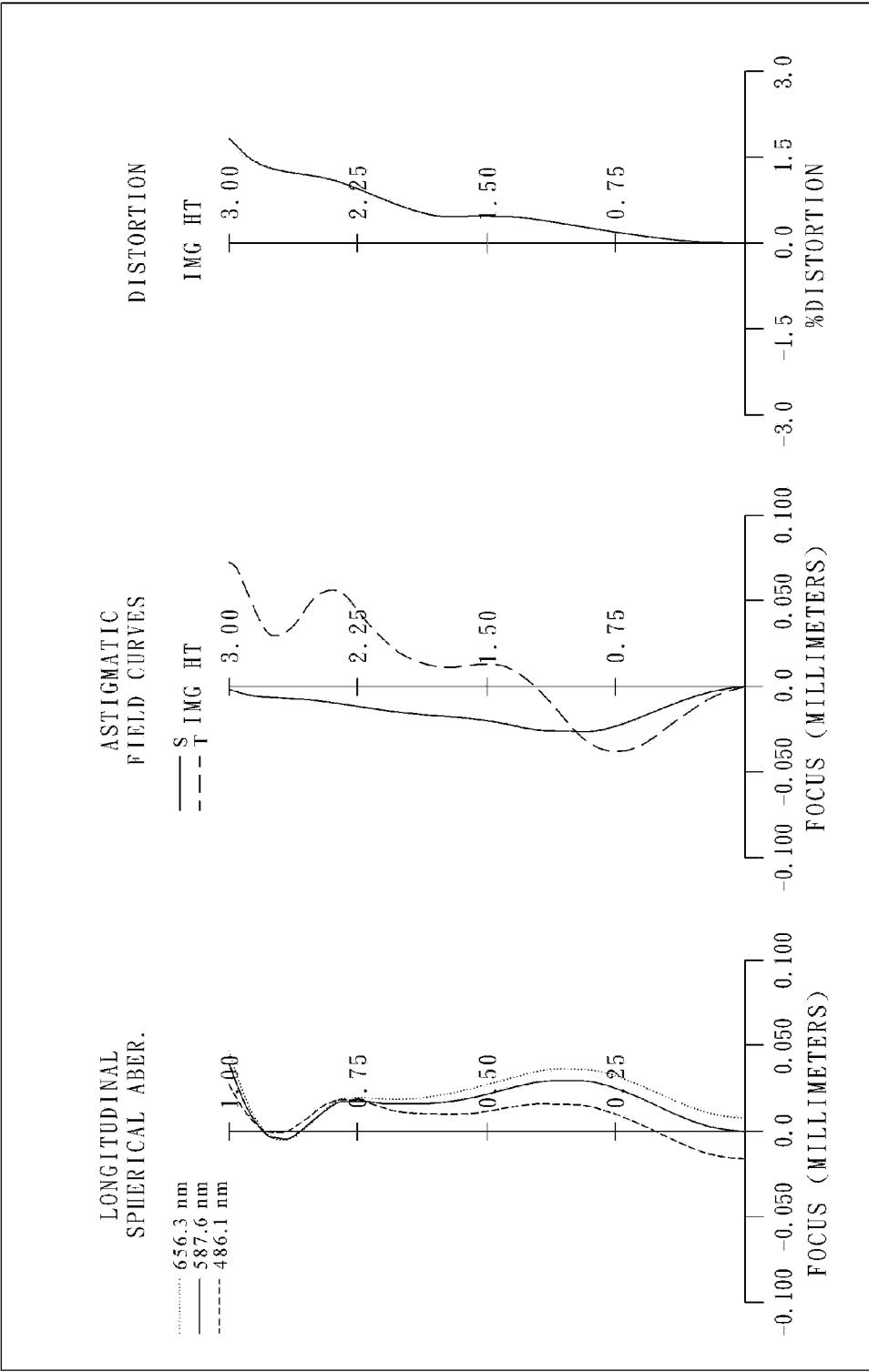


Fig. 6B

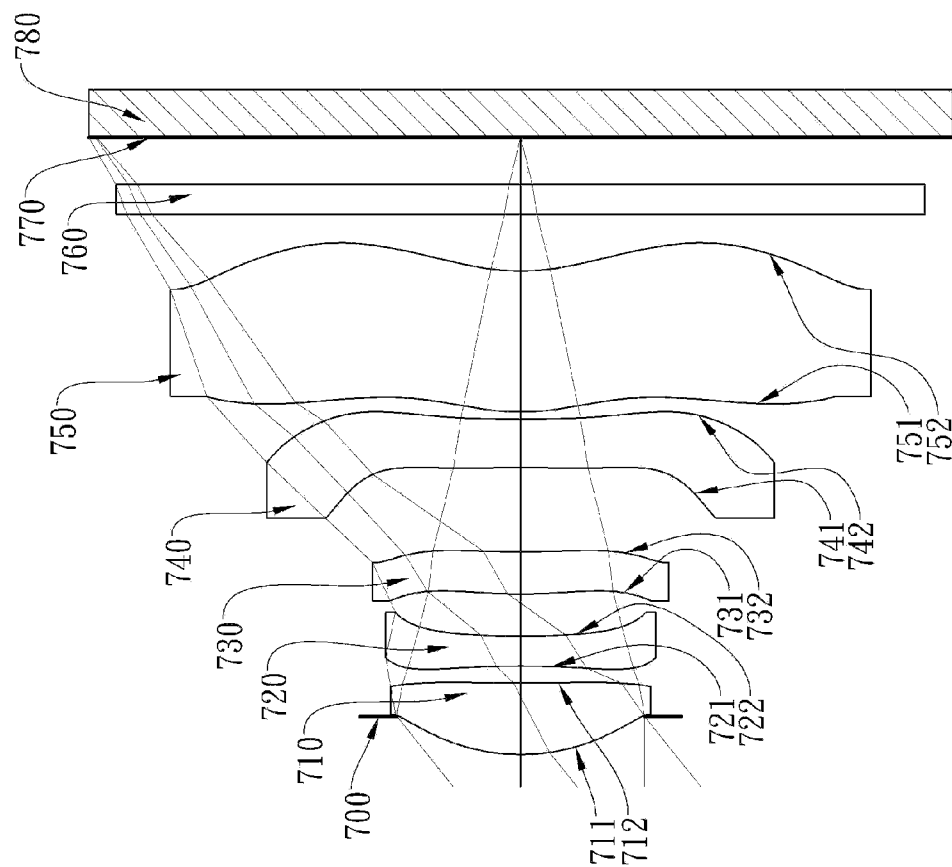


Fig. 7A

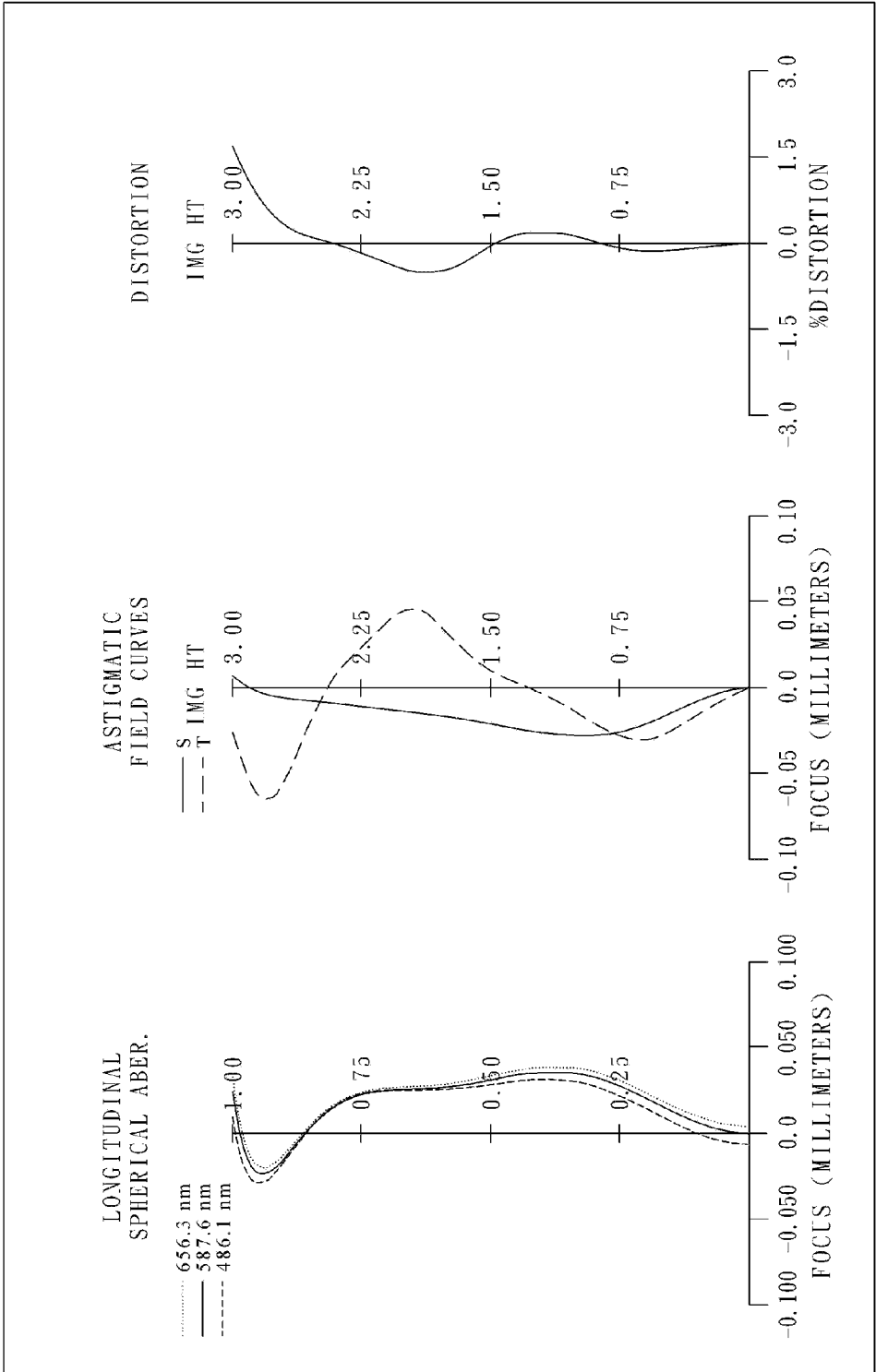


Fig. 7B

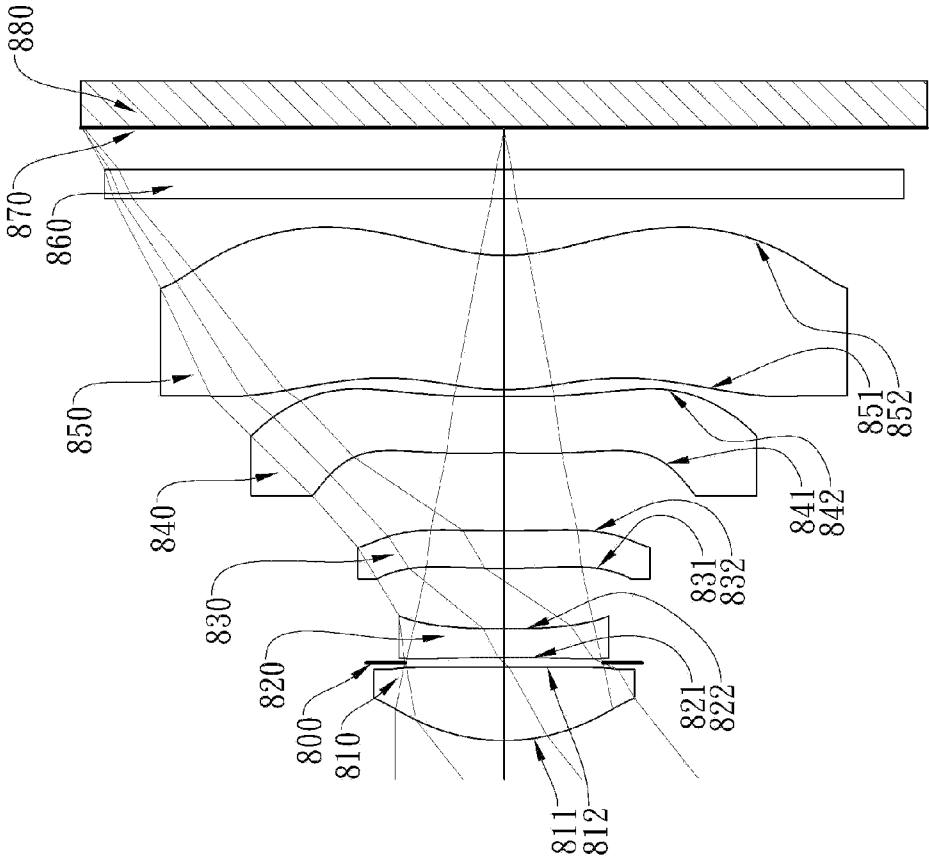


Fig. 8A

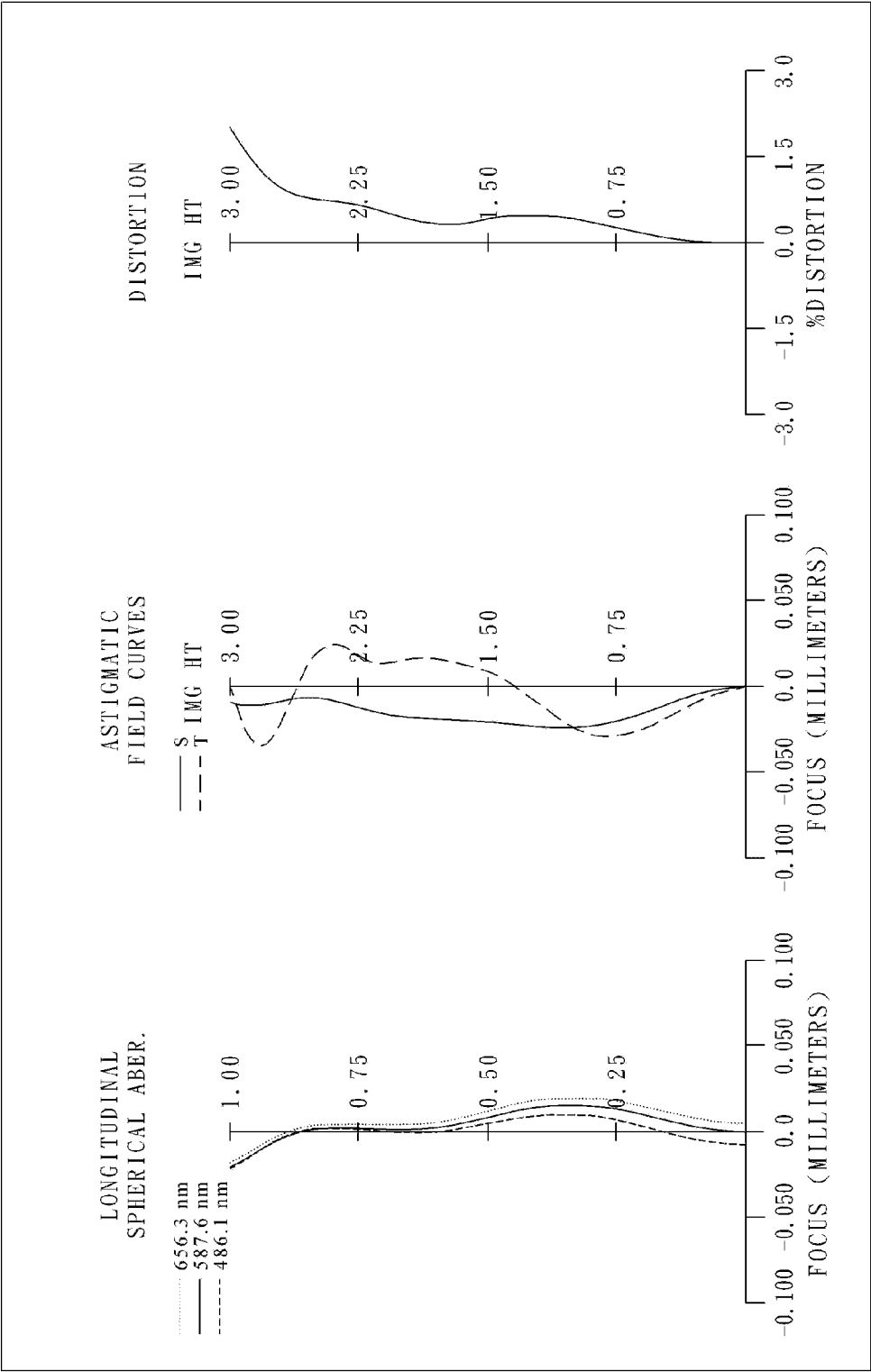


Fig. 8B

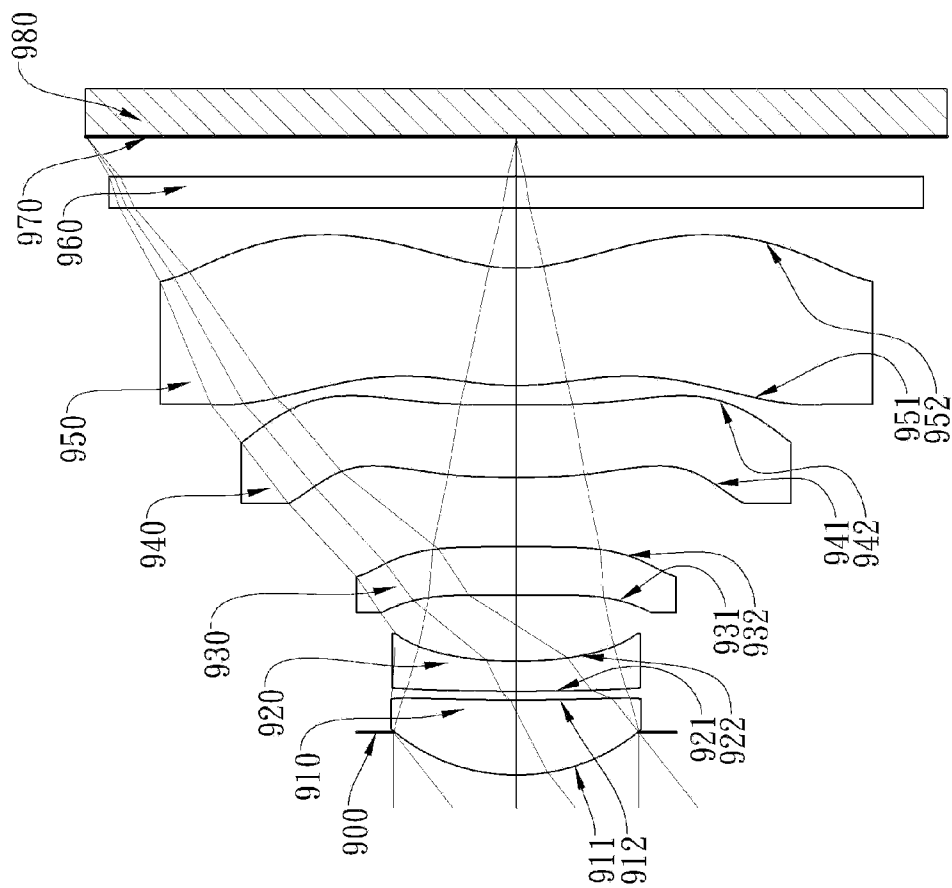


Fig. 9A

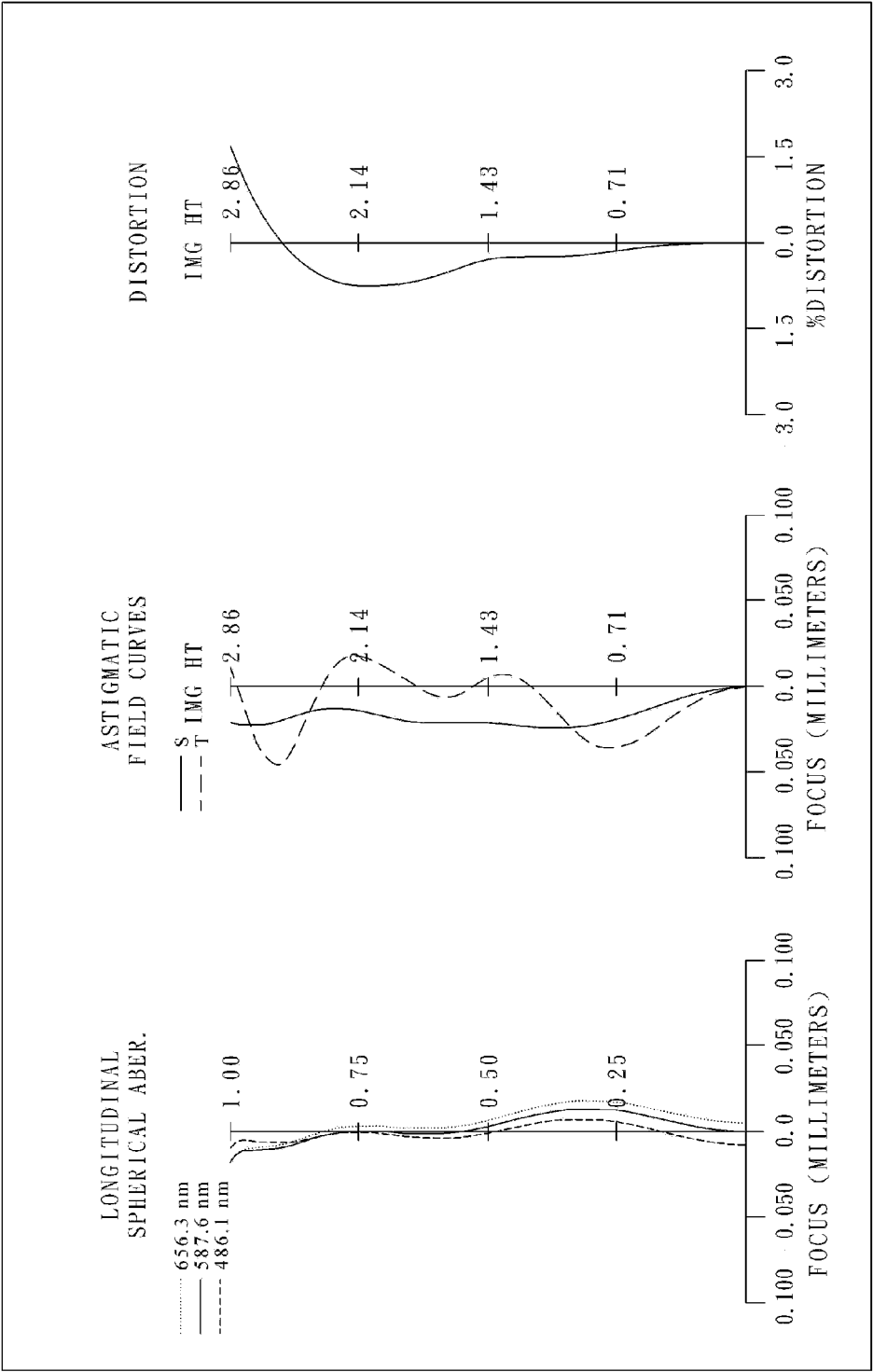


Fig. 9B

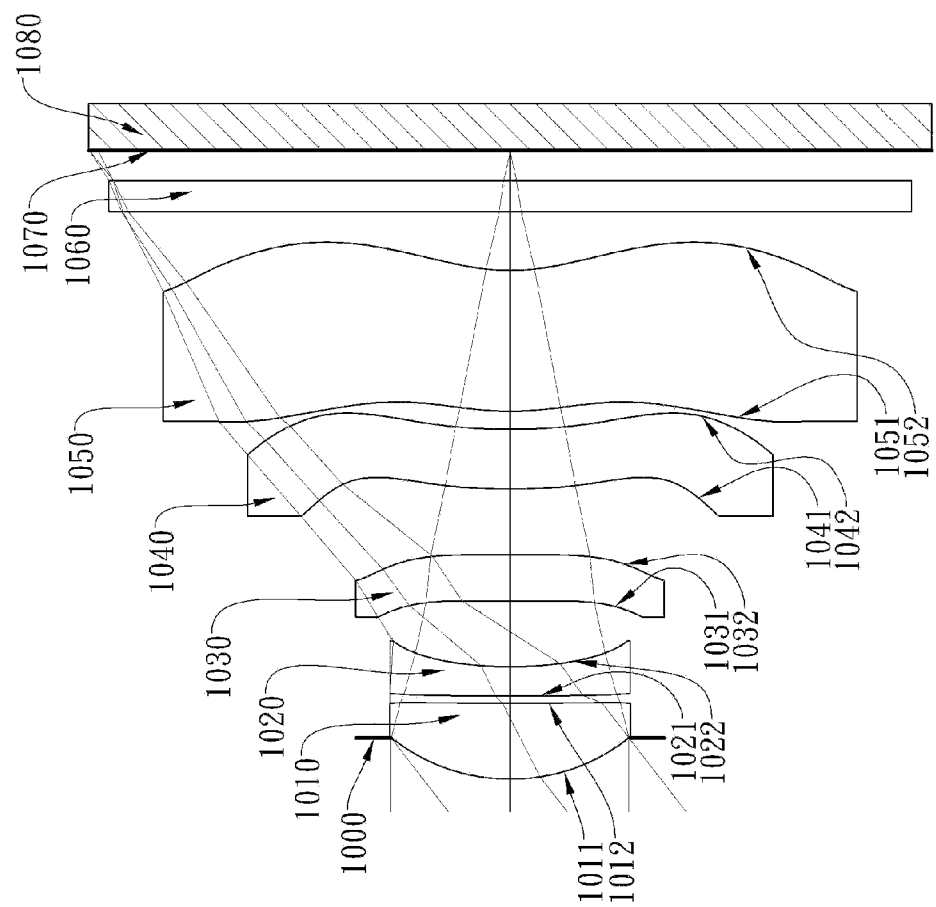


Fig. 10A

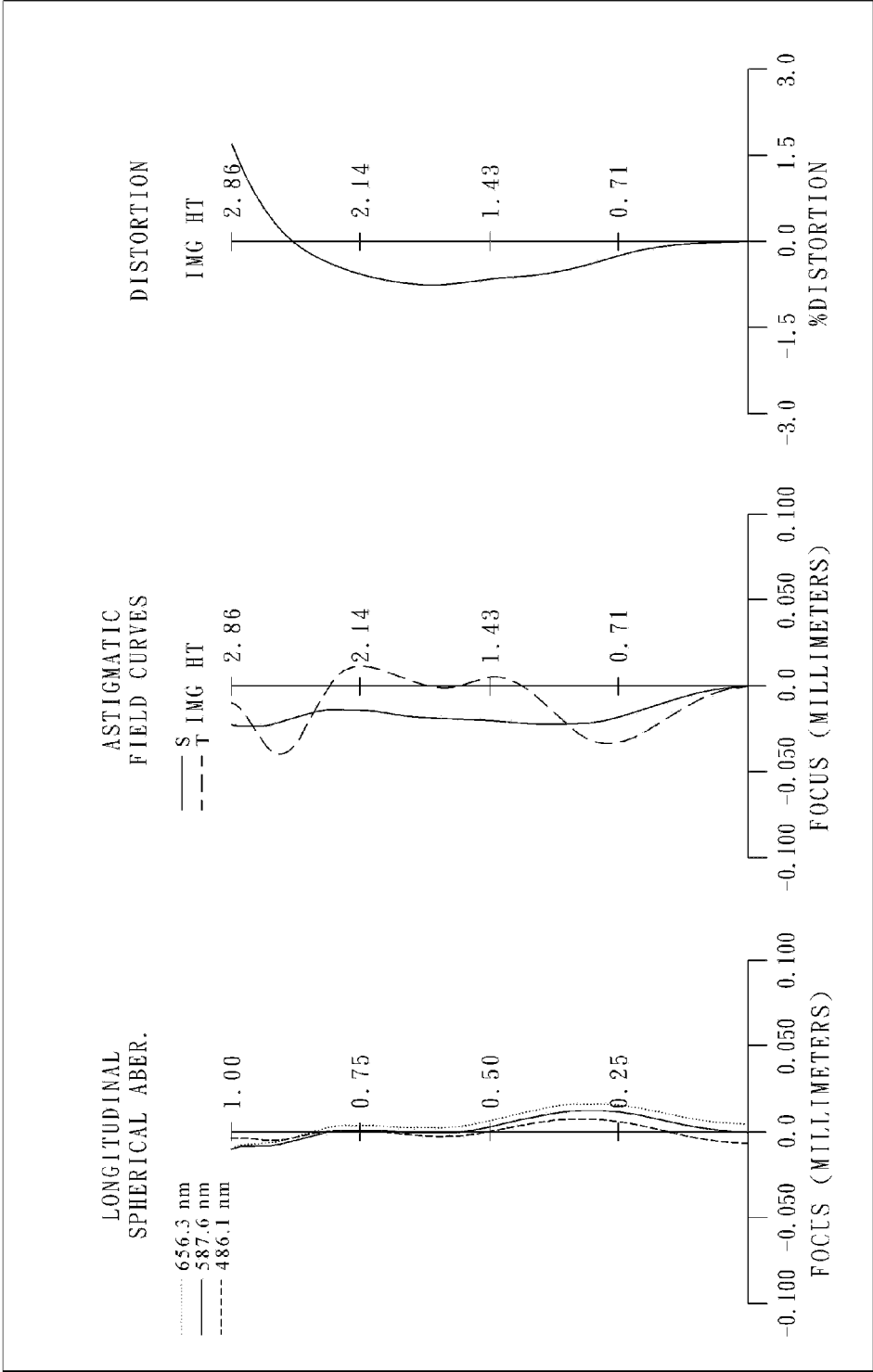


Fig. 10B

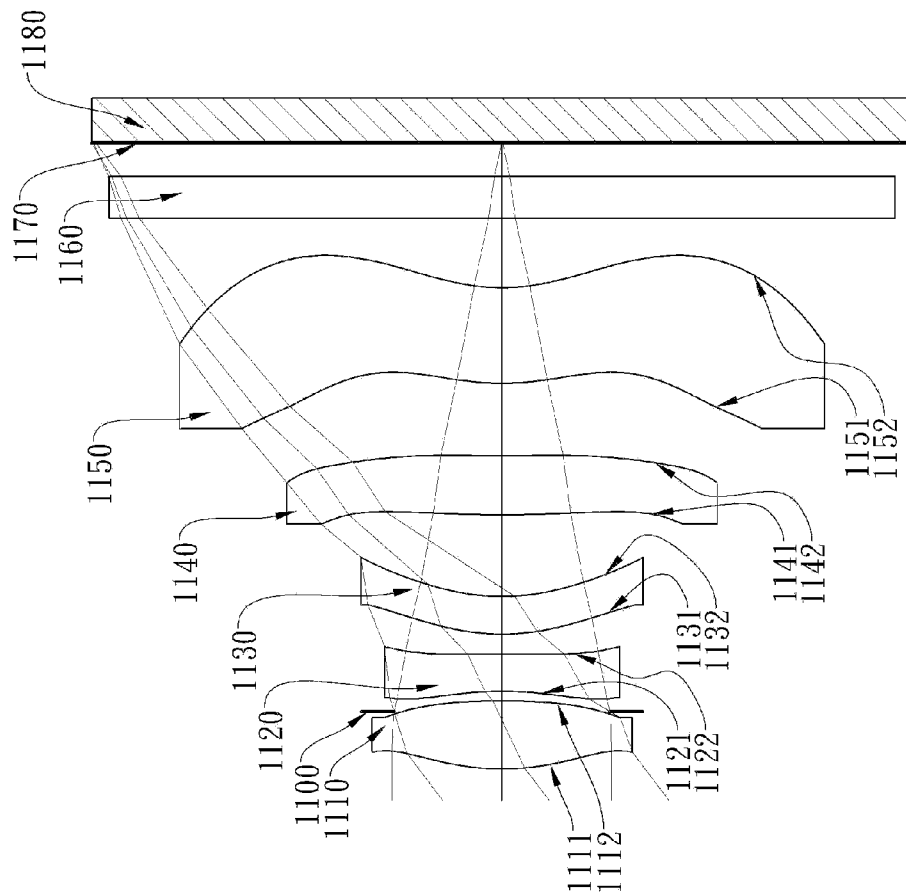


Fig. 11A

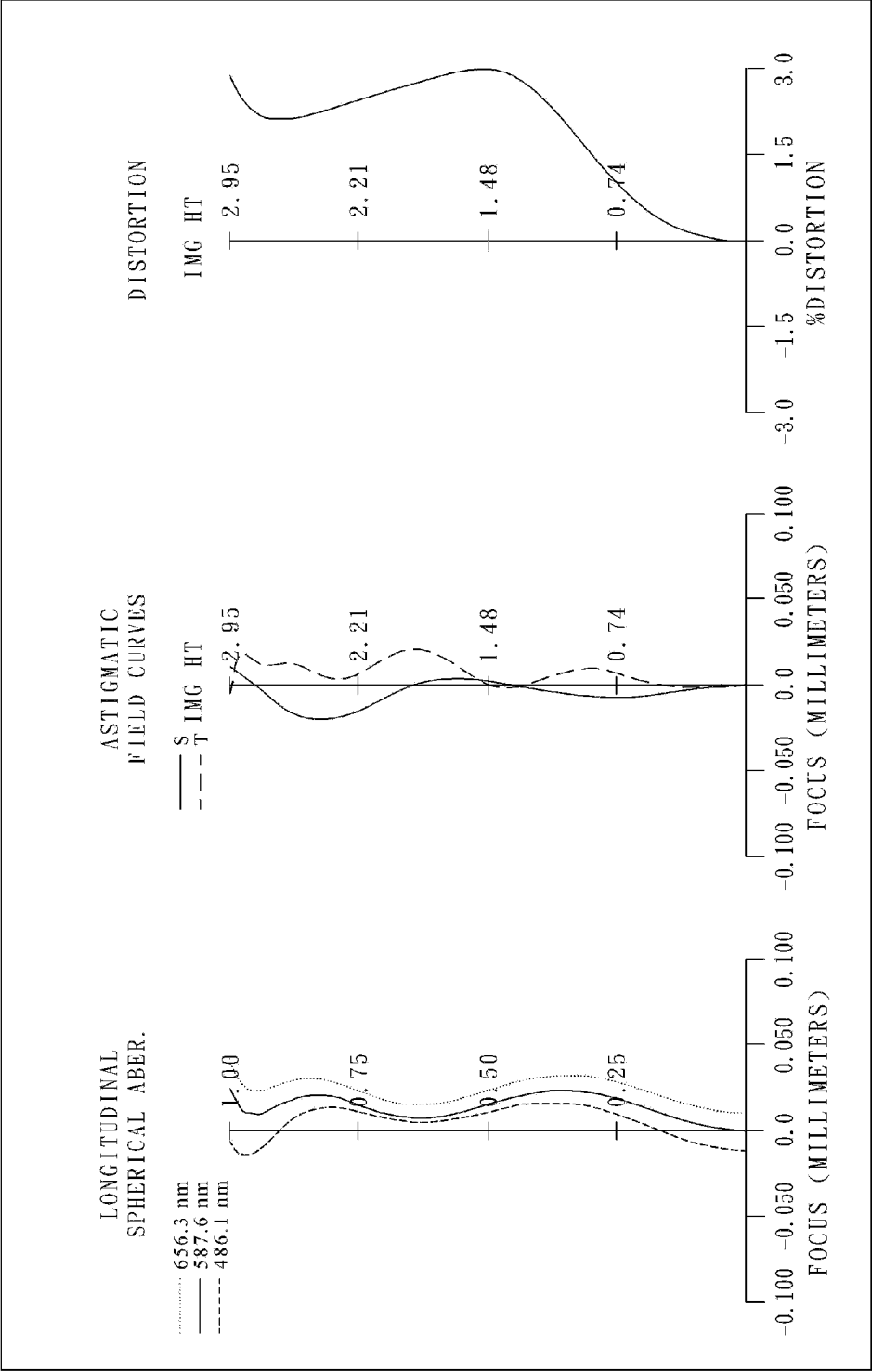


Fig. 11B

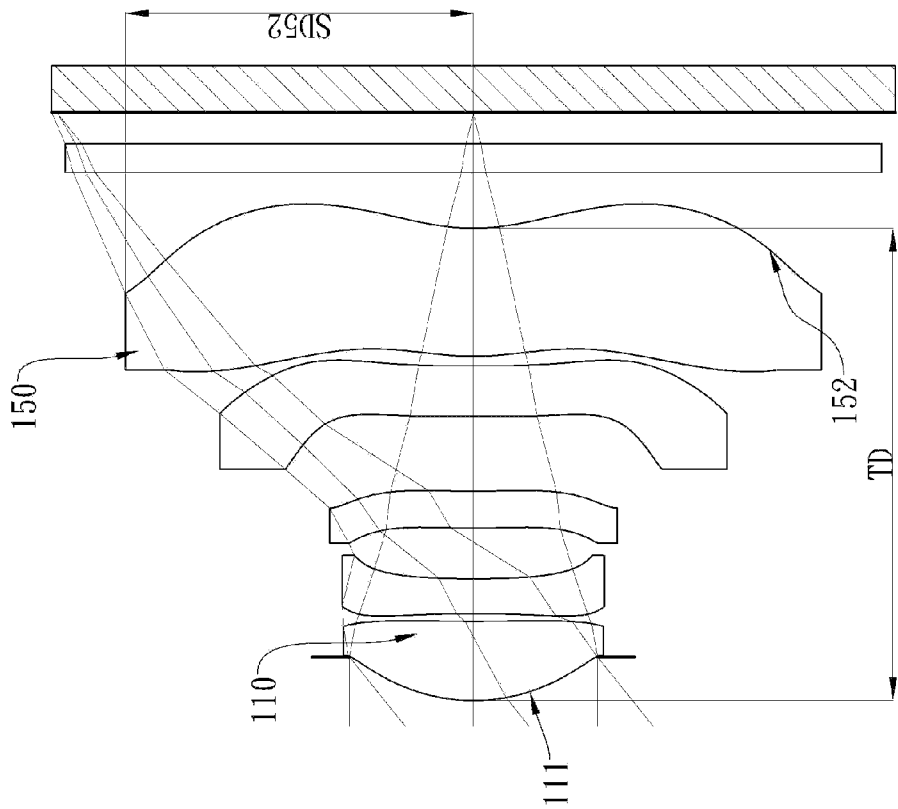


Fig. 12

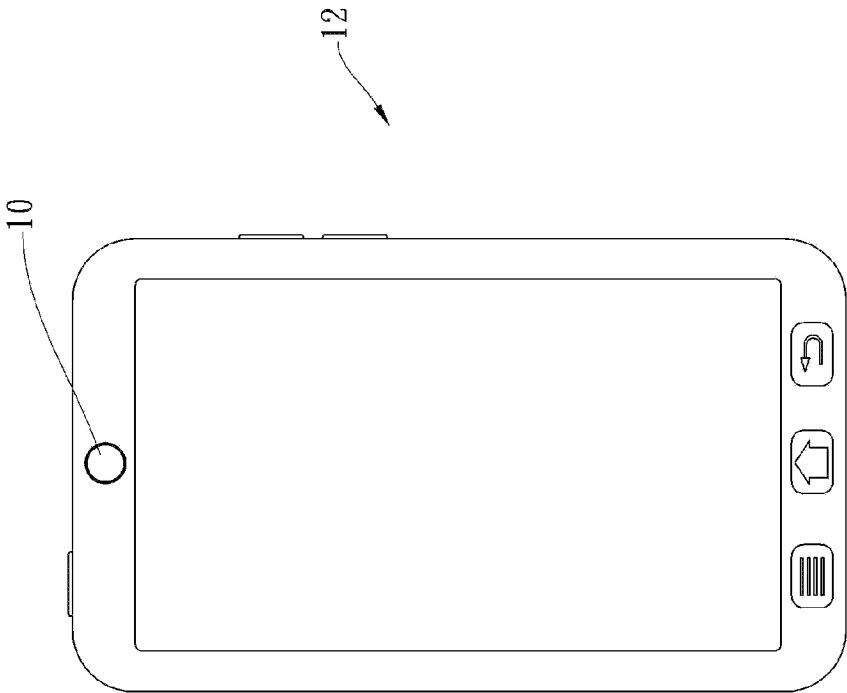


Fig. 13A

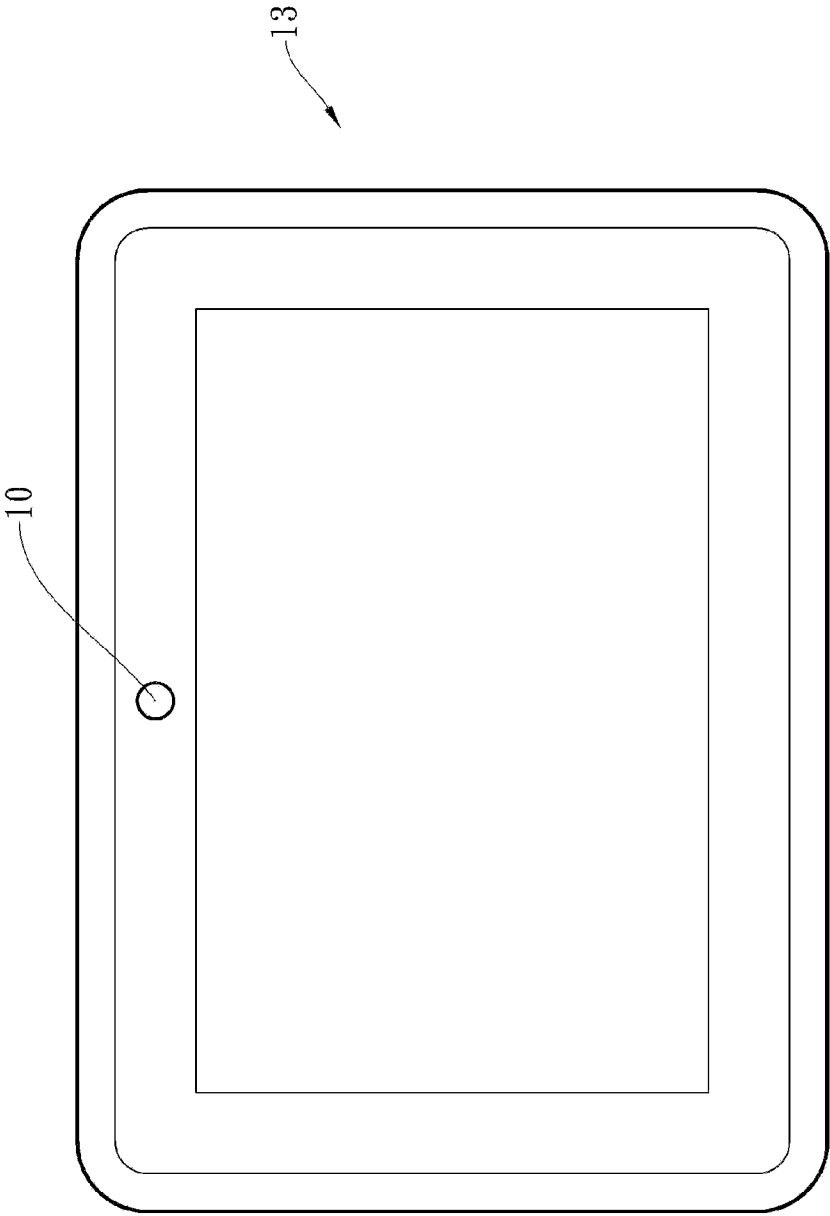


Fig. 13B

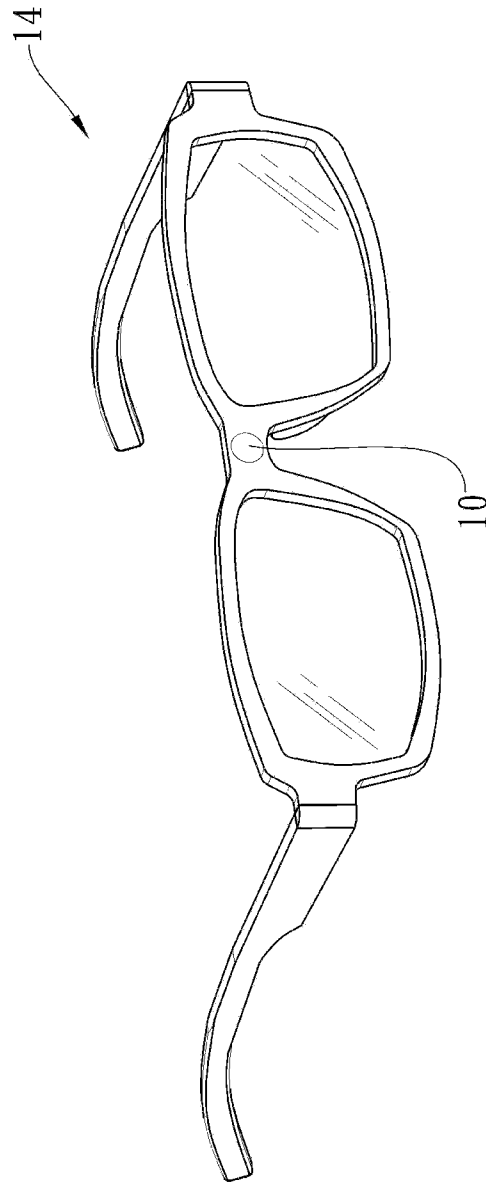


Fig. 13C

1

IMAGING OPTICAL LENS SYSTEM, IMAGING DEVICE AND MOBILE TERMINAL

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 102146841, filed Dec. 18, 2013, which is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to an imaging optical lens system, imaging device and mobile terminal. More particularly, the present disclosure relates to a compact imaging optical lens system and imaging device applicable to a mobile terminal.

2. Description of Related Art

In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

A conventional optical system employed in a portable electronic product mainly adopts a four-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. The conventional optical systems cannot satisfy these requirements of the compact optical systems.

Although other conventional optical systems with five-element lens structure enhance image quality, these are not favorable for correcting Petzval Sum. Moreover, it tends to produce excessive aberration in a peripheral region of an image which might cause vagueness and distortion in the peripheral region of the image.

In light of the foregoing, a need exists in the art for having an optical system favorable for effectively correcting the Petzval Sum. In addition, it might be favorable for reducing the aberration in the peripheral region of the system through adjusting the curvature of the image-side surface of the fourth lens element in order to avoid vagueness and distortion in the peripheral region

SUMMARY

According to one aspect of the present disclosure, an imaging optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The fourth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The fifth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein both of an object-side surface and the image-side surface of the fifth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the

2

fifth lens element has at least one inflection point. The imaging optical lens system has a total of five lens elements with refractive power. When a focal length of the imaging optical lens system is f , a curvature radius of the image-side surface of the fourth lens element is $R8$, a central thickness of the fifth lens element is $CT5$, and a central thickness of the fourth lens element is $CT4$, the following conditions are satisfied:

$$0 < f/R8 < 2.0; \text{ and}$$

$$1.3 < CT5/CT4 < 3.5.$$

According to another aspect of the present disclosure, an imaging device includes the imaging optical lens system according to the foregoing aspect and an image sensor.

According to still another aspect of the present disclosure, a mobile terminal includes the imaging device according to the foregoing aspect.

According to yet another aspect of the present disclosure, an imaging optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The fourth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein both of an object-side surface and the image-side surface of the fourth lens element are aspheric and at least one of the object-side surface and the image-side surface of the fourth lens element has at least one inflection point. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein both of the object-side surface and the image-side surface of the fifth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point. The imaging optical lens system has a total of five lens elements with refractive power. When a focal length of the imaging optical lens system is f , a curvature radius of the image-side surface of the fourth lens element is $R8$, a focal length of the second lens element is $f2$, a focal length of the fourth lens element is $f4$, and a focal length of the fifth lens element is $f5$, the following conditions are satisfied:

$$0 < f/R8 < 1.2;$$

$$|f/f2| < 0.8; \text{ and}$$

$$-3 < f4/f5 < 1.$$

According to still yet another aspect of the present disclosure, an imaging device includes the imaging optical lens system according to the foregoing aspect and an image sensor.

According to a further aspect of the present disclosure, a mobile terminal includes the imaging device according to the foregoing aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

FIG. 1A is a schematic view of an imaging device according to the 1st embodiment of the present disclosure;

3

FIG. 1B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 1st embodiment;

FIG. 2A is a schematic view of an imaging device according to the 2nd embodiment of the present disclosure;

FIG. 2B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 2nd embodiment;

FIG. 3A is a schematic view of an imaging device according to the 3rd embodiment of the present disclosure;

FIG. 3B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 3rd embodiment;

FIG. 4A is a schematic view of an imaging device according to the 4th embodiment of the present disclosure;

FIG. 4B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 4th embodiment;

FIG. 5A is a schematic view of an imaging device according to the 5th embodiment of the present disclosure;

FIG. 5B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 5th embodiment;

FIG. 6A is a schematic view of an imaging device according to the 6th embodiment of the present disclosure;

FIG. 6B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 6th embodiment;

FIG. 7A is a schematic view of an imaging device according to the 7th embodiment of the present disclosure;

FIG. 7B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 7th embodiment;

FIG. 8A is a schematic view of an imaging device according to the 8th embodiment of the present disclosure;

FIG. 8B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 8th embodiment;

FIG. 9A is a schematic view of an imaging device according to the 9th embodiment of the present disclosure;

FIG. 9B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 9th embodiment;

FIG. 10A is a schematic view of an imaging device according to the 10th embodiment of the present disclosure;

FIG. 10B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 10th embodiment;

FIG. 11A is a schematic view of an imaging device according to the 11th embodiment of the present disclosure;

FIG. 11B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 11th embodiment;

FIG. 12 shows TD and SD52 of the present disclosure according to the FIG. 1A;

FIG. 13A shows a smart phone with an imaging device of the present disclosure installed therein;

FIG. 13B shows a tablet personal computer with an imaging device of the present disclosure installed therein; and

FIG. 13C shows a wearable device with an imaging device of the present disclosure installed therein.

DETAILED DESCRIPTION

An imaging optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth

4

lens element. The imaging optical lens system has a total of five lens elements with refractive power.

The first lens element has positive refractive power, so that it provides the imaging optical lens system with the positive refractive power as it needs to be. The first lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for reducing the astigmatism.

The second lens element can have negative refractive power, so that it is favorable for correcting the aberration of the first lens element.

The third lens element has positive refractive power, so that it is favorable for reducing the system sensitivity. The third lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for effectively correcting the astigmatism.

The fourth lens element can have an object-side surface being concave in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein at least one of the object-side surface and the image-side surface of the fourth lens element can have at least one inflection point. Therefore, it is favorable for further correcting the aberration and reducing the incident angle of the light projecting onto an image sensor so as to increase the responding efficiency of the image sensor.

The fifth lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for further correcting the astigmatism. The object-side surface of the fifth lens element can be convex in a paraxial region thereof and change to a concave shape and then a convex shape from the paraxial region to a peripheral region. Moreover, at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point. Therefore, it is favorable for correcting aberration of the off-axis.

When a focal length of the imaging optical lens system is f , and a curvature radius of the image-side surface of the fourth lens element is $R8$, the following condition is satisfied: $0 < f/R8 < 2.0$. Therefore, it is favorable for correcting the aberration of the imaging optical lens system in a peripheral region by adjusting the curvature of the image-side surface of the fourth lens element. Preferably, the following condition is satisfied: $0 < f/R8 < 1.2$.

When a central thickness of the fifth lens element is $CT5$, and a central thickness of the fourth lens element is $CT4$, the following condition is satisfied: $1.3 < CT5/CT4 < 3.5$. Therefore, it is favorable for arranging the thicknesses of the imaging optical lens system so as to avoid the lens elements from being deformed. Preferably, the following condition is satisfied: $1.5 < CT5/CT4 < 3.0$.

When the focal length of the imaging optical lens system is f , and a focal length of the second lens element is $f2$, the following condition is satisfied: $1/f2 < 0.8$. Therefore, it is favorable for correcting the aberration of the first lens element.

When a focal length of the fourth lens element is $f4$, and a focal length of the fifth lens element is $f5$, the following condition is satisfied: $-3 < f4/f5 < 1$. Therefore, it is favorable for balancing the refractive powers of the lens elements so as to reduce the sensitivity of the imaging optical lens system.

When an axial distance between the object-side surface of the first lens element and the image-side surface of the fifth lens element is TD , and a vertical distance between a maximum effective radius position on the image-side surface of

5

the fifth lens element and an optical axis is SD52, the following condition is satisfied: $0.50 < TD/(2 \cdot SD52) < 0.80$. Therefore, it is favorable for reducing the space required for assembling the lens elements so as to keep the imaging optical lens system more compact.

When an axial distance between the object-side surface of the first lens element and the image plane is TL, and a maximum image height of the imaging optical lens system (half of a diagonal length of an effective photosensitive area of an image sensor) is ImgH, the following condition is satisfied: $TL/ImgH < 1.7$. Therefore, it is favorable for keeping the imaging optical lens system compact so as to be applied to compact and portable electronic products.

When a curvature radius of the image-side surface of the fifth lens element is R10, and a curvature radius of the image-side surface of the third lens element is R6, the following condition is satisfied: $0 < R10/R6 < 1.0$. Therefore, it is favorable for effectively correcting the Petzval Sum. Preferably, the following condition is satisfied: $0 < R10/R6 < 0.5$.

When the focal length of the imaging optical lens system is f, and the focal length of the fourth lens element is f4, the following condition is satisfied: $-0.7 < f/f4 < 0.3$. Therefore, it is favorable for further correcting the aberration. Preferably, the following condition is satisfied: $-0.6 < f/f4 < 0$.

When an axial distance between the second lens element and the third lens element is T23, and a central thickness of the third lens element is CT3, the following condition is satisfied: $0.2 < CT3/T23 < 3.0$. Therefore, it is favorable for assembling the lens elements so as to improve the manufacturing yield rate.

When an axial distance between the first lens element and the second lens element is T12, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the second lens element and the third lens element is T23, and an axial distance between the third lens element and the fourth lens element is T34, the following condition is satisfied: $0 < (T12 + T45)/(T23 + T34) < 0.40$. Therefore, it is favorable for properly adjusting the axial distance between every lens elements so as to keep the imaging optical lens system compact.

When an Abbe number of the second lens element is V2, and an Abbe number of the fourth lens element is V4, the following condition is satisfied: $0.8 < V2/V4 < 1.3$. Therefore, it is favorable for effectively correcting the chromatic aberration of the imaging optical lens system.

When the focal length of the imaging optical lens system is f, a focal length of the third lens element is f3, and the focal length of the fifth lens element is f5, the following condition is satisfied: $0 < |f/f3| + |f/f5| < 0.5$. Therefore, it is favorable for balancing the arrangement of the refractive powers so as to keep the imaging optical lens system compact and reduce the sensitivity thereof.

According to the imaging optical lens system of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the imaging optical lens system may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the imaging optical lens system can also be reduced.

6

According to the imaging optical lens system of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof.

According to the imaging optical lens system of the present disclosure, the imaging optical lens system can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the imaging optical lens system of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the imaging optical lens system and the image plane and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image plane is favorable for enlarging the field of view of the imaging optical lens system and thereby provides a wider field of view for the same.

The present imaging optical lens system can be optionally applied to moving focus or zoom optical systems. According to the imaging optical lens system of the present disclosure, the imaging optical lens system is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals.

According to the present disclosure, an imaging device is provided. The imaging device includes the imaging optical lens system according to the foregoing imaging optical lens system of the present disclosure, and an image sensor, wherein the image sensor is disposed on an image plane of the foregoing imaging optical lens system. Therefore, the imaging device can obtain best image quality through the design of the imaging optical lens system. Preferably, the imaging device can further include a barrel member, a holding member or a combination thereof.

In FIG. 13A, FIG. 13B and FIG. 13C, an imaging device 10 may be installed in but not limited to a mobile terminal, including a smart phone 12, a tablet personal computer 13 or a wearable device 14. The three exemplary figures of different kinds of mobile terminal are only exemplary for showing the imaging device of present disclosure installing in a mobile terminal and is not limited thereto. Preferably, the mobile terminal can further include but not limited to display, control unit, random access memory unit (RAM) a read only memory unit (ROM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-11th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1A is a schematic view of an imaging device according to the 1st embodiment of the present disclosure. FIG. 1B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 1st embodiment.

In FIG. 1A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor **180**. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop **100**, a first lens element **110**, a second lens element **120**, a third lens element **130**, a fourth lens element **140**, a fifth lens element **150**, an IR-cut filter **160** and an image plane **170**, wherein the imaging optical lens system has a total of five lens elements (**110-150**) with refractive power.

The first lens element **110** with positive refractive power has an object-side surface **111** being convex in a paraxial region thereof and an image-side surface **112** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **111** and the image-side surface **112** of the first lens element **110** are aspheric.

The second lens element **120** with negative refractive power has an object-side surface **121** being concave in a paraxial region thereof and an image-side surface **122** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **121** and the image-side surface **122** of the second lens element **120** are aspheric.

The third lens element **130** with positive refractive power has an object-side surface **131** being convex in a paraxial region thereof and an image-side surface **132** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **131** and the image-side surface **132** of the third lens element **130** are aspheric.

The fourth lens element **140** with negative refractive power has an object-side surface **141** being concave in a paraxial region thereof and an image-side surface **142** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **141** and the image-side surface **142** of the fourth lens element **140** are aspheric. Furthermore, both of the object-side surface **141** and the image-side surface **142** of the fourth lens element **140** have at least one inflection point.

The fifth lens element **150** with negative refractive power has an object-side surface **151** being convex in a paraxial region thereof and an image-side surface **152** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **151** and the image-side surface **152** of the fifth lens element **150** are aspheric. Furthermore, both of the object-side surface **151** and the image-side surface **152** of the fifth lens element **150** have at least one inflection point, wherein the object-side surface **151** of the fifth lens element **150** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **160** is made of glass and located between the fifth lens element **150** and the image plane **170**, and will not affect the focal length of the imaging optical lens system. The image sensor **180** is disposed on the image plane **170** of the imaging optical lens system.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (A_i \times (Y^i)),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

A_i is the i-th aspheric coefficient.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when a focal length of the imaging optical lens system is f, an f-number of the imaging optical lens system is Fno, and half of a maximal field of view of the imaging optical lens system is HFOV, these parameters have the following values: f=3.66 mm; Fno=2.05; and HFOV=38.8 degrees.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when an Abbe number of the second lens element **120** is V2, and an Abbe number of the fourth lens element **140** is V4, the following condition is satisfied: V2/V4=1.00.

In the imaging optical lens system according to the 1st embodiment, a central thickness of the third lens element **130** is CT3, and an axial distance between the second lens element **120** and the third lens element **130** is T23, the following condition is satisfied: CT3/T23=0.73.

In the imaging optical lens system according to the 1st embodiment, a central thickness of the fourth lens element **140** is CT4, and a central thickness of the fifth lens element **150** is CT5, the following condition is satisfied: CT5/CT4=2.54.

In the imaging optical lens system according to the 1st embodiment, an axial distance between the first lens element **110** and the second lens element **120** is T12, the axial distance between the second lens element **120** and the third lens element **130** is T23, an axial distance between the third lens element **130** and the fourth lens element **140** is T34, and an axial distance between the fourth lens element **140** and the fifth lens element **150** is T45, the following condition is satisfied: (T12+T45)/(T23+T34)=0.14.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when a curvature radius of the image-side surface **152** of the fifth lens element **150** is R10, and a curvature radius of the image-side surface **132** of the third lens element **130** is R6, the following condition is satisfied: R10/R6=0.15.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when the focal length of the imaging optical lens system is f, and a curvature radius of the image-side surface **142** of the fourth lens element **140** is R8, the following condition is satisfied: f/R8=0.12.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when the focal length of the imaging optical lens system is f, and a focal length of the second lens element **120** is f2, the following condition is satisfied: f/f2=0.58.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when the focal length of the imaging optical lens system is f, and a focal length of the fourth lens element **140** is f4, the following condition is satisfied: f/f4=-0.27.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when the focal length of the fourth lens element **140** is f4, and a focal length of the fifth lens element **150** is f5, the following condition is satisfied: f4/f5=0.45.

In the imaging optical lens system of the imaging device according to the 1st embodiment, when the focal length of the imaging optical lens system is f, a focal length of the third lens

element **130** is f3, and the focal length of the fifth lens element **150** is f5, the following condition is satisfied: $|f/f3|+|f/f5|=0.20$.

In the imaging optical lens system according to the 1st embodiment, when an axial distance between the object-side surface **111** of the first lens element **110** and the image-side surface **152** of the fifth lens element **150** is TD, and a vertical distance between a maximum effective radius position on the image-side surface **152** of the fifth lens element **150** and an optical axis is SD52, the following condition is satisfied: $TD/(2*SD52)=0.68$.

In the imaging optical lens system according to the 1st embodiment, when an axial distance between the object-side surface **111** of the first lens element **110** and the image plane **170** is TL, and a maximum image height of the imaging optical lens system (half of a diagonal length of an effective photosensitive area of the image sensor **180**) is ImgH, the following condition is satisfied: $TL/ImgH=1.41$.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment							
f = 3.66 mm, Fno = 2.05, HFOV = 38.8 deg.							
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe # Focal Length
0	Object	Plano		Infinity			
1	Ape. Stop	Plano		-0.313			
2	Lens 1	1.311	ASP	0.570	Plastic	1.544	55.9 2.66
3		11.816	ASP	0.055			
4	Lens 2	-11.228	ASP	0.251	Plastic	1.640	23.3 -6.34
5		6.402	ASP	0.366			
6	Lens 3	7.335	ASP	0.266	Plastic	1.640	23.3 44.65
7		9.729	ASP	0.540			
8	Lens 4	-12.373	ASP	0.363	Plastic	1.640	23.3 -13.78
9		31.023	ASP	0.070			
10	Lens 5	1.998	ASP	0.922	Plastic	1.544	55.9 -30.88
11		1.496	ASP	0.400			
12	IR-cut filter	Plano		0.210	Glass	1.517	64.2 —
13		Plano		0.225			
14	Image	Plano		—			

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	9.8077E-03	-1.0000E+00	4.7208E+00	1.8728E+00	-1.8917E+01
A4 =	-2.3323E-02	-7.7742E-02	-6.0424E-02	1.8926E-02	-1.7775E-01
A6 =	1.1870E-01	-2.5842E-01	4.6781E-02	2.1114E-01	-2.5421E-01
A8 =	-3.7649E-01	1.1929E+00	7.5647E-01	-1.7898E-02	-8.6590E-02
A10 =	5.2433E-01	-2.2539E+00	-1.8375E+00	3.2432E-02	3.1521E+00
A12 =	-3.9212E-01	1.9895E+00	1.9821E+00	-2.8787E-01	-9.0201E+00
A14 =	2.8611E-02	-7.0385E-01	-7.4115E-01	5.3567E-01	1.0258E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	-2.4928E+01	-1.7129E+01	-1.0000E+00	-1.0880E+00	-3.6899E+00
A4 =	-1.5324E-01	3.6368E-01	1.5016E-01	-3.8828E-01	-1.7443E-01
A6 =	7.3889E-03	-7.4219E-01	-2.0451E-01	2.5343E-01	1.0880E-01
A8 =	-8.7548E-01	7.9032E-01	1.2070E-01	-1.0126E-01	-5.4300E-02
A10 =	2.9868E+00	-6.0328E-01	-5.8364E-02	2.6082E-02	1.7585E-02
A12 =	-4.5338E+00	2.6133E-01	2.0881E-02	-4.1733E-03	-3.3680E-03
A14 =	3.3158E+00	-6.0577E-02	-4.1847E-03	3.7690E-04	3.4142E-04
A16 =	-9.1257E-01	7.0919E-03	3.3573E-04	-1.4698E-05	-1.3985E-05

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-14 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. This information related to Table 1 and Table 2 applies also to the Tables for the remaining embodiments, and so an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 2A is a schematic view of an imaging device according to the 2nd embodiment of the present disclosure. FIG. 2B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 2nd embodiment.

In FIG. 2A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor **280**. The imaging optical lens

11

system includes, in order from an object side to an image side, an aperture stop **200**, a first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens element **250**, an IR-cut filter **260** and an image plane **270**, wherein the imaging optical lens system has a total of five lens elements (**210-250**) with refractive power.

The first lens element **210** with positive refractive power has an object-side surface **211** being convex in a paraxial region thereof and an image-side surface **212** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **211** and the image-side surface **212** of the first lens element **210** are aspheric.

The second lens element **220** with negative refractive power has an object-side surface **221** being concave in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **221** and the image-side surface **222** of the second lens element **220** are aspheric.

The third lens element **230** with positive refractive power has an object-side surface **231** being convex in a paraxial region thereof and an image-side surface **232** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **231** and the image-side surface **232** of the third lens element **230** are aspheric.

The fourth lens element **240** with negative refractive power has an object-side surface **241** being convex in a paraxial region thereof and an image-side surface **242** being concave

12

in a paraxial region thereof, and is made of plastic material. The object-side surface **241** and the image-side surface **242** of the fourth lens element **240** are aspheric. Furthermore, both of the object-side surface **241** and the image-side surface **242** of the fourth lens element **240** have at least one inflection point.

The fifth lens element **250** with negative refractive power has an object-side surface **251** being convex in a paraxial region thereof and an image-side surface **252** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **251** and the image-side surface **252** of the fifth lens element **250** are aspheric. Furthermore, both of the object-side surface **251** and the image-side surface **252** of the fifth lens element **250** have at least one inflection point, wherein the object-side surface **251** of the fifth lens element **250** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **260** is made of glass and located between the fifth lens element **250** and the image plane **270**, and will not affect the focal length of the imaging optical lens system. The image sensor **280** is disposed on the image plane **270** of the imaging optical lens system.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment							
f = 3.34 mm, Fno = 2.00, HFOV = 39.9 deg.							
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe # Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.272				
2	Lens 1	1.323 ASP	0.505	Plastic	1.544	55.9	2.80
3		8.752 ASP	0.069				
4	Lens 2	-36.685 ASP	0.207	Plastic	1.650	21.4	-6.83
5		5.057 ASP	0.334				
6	Lens 3	9.479 ASP	0.319	Plastic	1.544	55.9	19.28
7		97.044 ASP	0.494				
8	Lens 4	99.149 ASP	0.365	Plastic	1.650	21.4	-16.84
9		9.841 ASP	0.100				
10	Lens 5	1.796 ASP	0.935	Plastic	1.544	55.9	-30.79
11		1.324 ASP	0.400				
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
13		Plano	0.147				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	1.2278E-02	-1.0000E+00	3.9434E+00	-7.0243E+00	4.9145E+00
A4 =	-2.3249E-02	-6.7598E-02	-7.0414E-02	1.3522E-02	-1.6297E-01
A6 =	1.1890E-01	-2.6739E-01	5.5379E-02	2.1564E-01	-2.3462E-01
A8 =	-3.8123E-01	1.1688E+00	7.5061E-01	-1.0965E-02	-1.2050E-01
A10 =	5.2550E-01	-2.2619E+00	-1.8221E+00	8.2631E-02	3.1392E+00
A12 =	-4.0021E-01	2.0020E+00	1.9956E+00	-2.8567E-01	-8.9885E+00
A14 =	5.5307E-03	-6.9293E-01	-7.4102E-01	4.7316E-01	1.0316E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	-1.0000E+00	-1.0000E+00	-1.0000E+00	-1.6052E+00	-3.4721E+00
A4 =	-1.2037E-01	3.5376E-01	1.3602E-01	-3.9726E-01	-1.6768E-01

TABLE 4-continued

Aspheric Coefficients					
A6 =	-3.0435E-02	-7.3442E-01	-2.0133E-01	2.5479E-01	1.0858E-01
A8 =	-8.7496E-01	7.8973E-01	1.1783E-01	-1.0100E-01	-5.4354E-02
A10 =	2.9962E+00	-6.0282E-01	-5.8386E-02	2.6111E-02	1.7591E-02
A12 =	-4.5265E+00	2.6201E-01	2.0957E-02	-4.1722E-03	-3.3643E-03
A14 =	3.3185E+00	-6.0729E-02	-4.1720E-03	3.7605E-04	3.4123E-04
A16 =	-9.0979E-01	6.8869E-03	3.3653E-04	-1.5041E-05	-1.4034E-05

In the imaging optical lens system according to the 2nd embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment. Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	3.34	f/R8	0.34
Fno	2.00	f/f2	0.49
HFOV [deg.]	39.9	f/f4	-0.20
V2/V4	1.00	f4/f5	0.55
CT3/T23	0.96	f/f3 + f/f5	0.28
CT5/CT4	2.56	TD/(2*SD52)	0.68
(T12 + T45)/(T23 + T34)	0.20	TL/lmgH	1.43
R10/R6	0.01		

3rd Embodiment

FIG. 3A is a schematic view of an imaging device according to the 3rd embodiment of the present disclosure. FIG. 3B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 3rd embodiment.

In FIG. 3A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor **380**. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop **300**, a first lens element **310**, a second lens element **320**, a third lens element **330**, a fourth lens element **340**, a fifth lens element **350**, an IR-cut filter **360** and an image plane **370**, wherein the imaging optical lens system has a total of five lens elements (**310-350**) with refractive power.

The first lens element **310** with positive refractive power has an object-side surface **311** being convex in a paraxial region thereof and an image-side surface **312** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **311** and the image-side surface **312** of the first lens element **310** are aspheric.

The second lens element **320** with negative refractive power has an object-side surface **321** being convex in a paraxial region thereof and an image-side surface **322** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **321** and the image-side surface **322** of the second lens element **320** are aspheric.

The third lens element **330** with positive refractive power has an object-side surface **331** being convex in a paraxial region thereof and an image-side surface **332** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **331** and the image-side surface **332** of the third lens element **330** are aspheric.

The fourth lens element **340** with negative refractive power has an object-side surface **341** being concave in a paraxial region thereof and an image-side surface **342** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **341** and the image-side surface **342** of the fourth lens element **340** are aspheric. Furthermore, both of the object-side surface **341** and the image-side surface **342** of the fourth lens element **340** have at least one inflection point.

The fifth lens element **350** with positive refractive power has an object-side surface **351** being convex in a paraxial region thereof and an image-side surface **352** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **351** and the image-side surface **352** of the fifth lens element **350** are aspheric. Furthermore, both of the object-side surface **351** and the image-side surface **352** of the fifth lens element **350** have at least one inflection point, wherein the object-side surface **351** of the fifth lens element **350** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **360** is made of glass and located between the fifth lens element **350** and the image plane **370**, and will not affect the focal length of the imaging optical lens system. The image sensor **380** is disposed on the image plane **370** of the imaging optical lens system.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment f = 4.38 mm, Fno = 2.20, HFOV = 38.2 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.340					
2	Lens 1	1.477 ASP	0.641	Plastic	1.565	56.5	3.69	
3		4.261 ASP	0.068					
4	Lens 2	88.658 ASP	0.298	Plastic	1.640	23.3	-17.51	
5		9.933 ASP	0.454					
6	Lens 3	15.519 ASP	0.277	Plastic	1.640	23.3	125.72	
7		19.096 ASP	0.614					
8	Lens 4	-9.343 ASP	0.454	Plastic	1.640	23.3	-11.17	

TABLE 5-continued

3rd Embodiment							
f = 4.38 mm, Fno = 2.20, HFOV = 38.2 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
9		31.023	ASP	0.070			
10	Lens 5	2.114	ASP	0.960	Plastic	1.544	55.9
11		1.791	ASP	0.467			
12	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
13		Plano	0.405				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	2.0278E-02	-1.0000E+00	-3.0000E+01	-2.1697E+01	-2.7593E+01
A4 =	-1.4081E-02	-7.7574E-02	-7.7381E-02	1.3829E-02	-1.1412E-01
A6 =	4.9139E-02	-1.6275E-01	2.2018E-02	1.3812E-01	-1.2949E-01
A8 =	-1.3212E-01	3.9487E-01	2.6434E-01	2.2852E-02	-3.7855E-02
A10 =	1.3728E-01	-5.4528E-01	-4.5453E-01	-7.8202E-03	8.0308E-01
A12 =	-6.0392E-02	3.8516E-01	3.6349E-01	-5.5523E-02	-1.6451E+00
A14 =	-1.5846E-02	-1.1345E-01	-1.0088E-01	8.8655E-02	1.3638E+00
A16 =					-4.1265E-01
Surface #	7	8	9	10	11
k =	-3.0000E+01	-2.9883E+01	-1.0000E+00	-2.4083E+00	-3.9058E+00
A4 =	-1.0390E-01	2.1898E-01	8.0897E-02	-2.4569E-01	-1.1890E-01
A6 =	-6.9892E-03	-3.4318E-01	-8.9792E-02	1.1766E-01	5.1263E-02
A8 =	-2.9606E-01	2.6654E-01	4.0883E-02	-3.4367E-02	-1.8462E-02
A10 =	7.4589E-01	-1.5062E-01	-1.4707E-02	6.5144E-03	4.3844E-03
A12 =	-8.3227E-01	4.7942E-02	3.8281E-03	-7.6579E-04	-6.1857E-04
A14 =	4.4668E-01	-8.1040E-03	-5.6234E-04	5.0754E-05	4.6043E-05
A16 =	-9.0324E-02	7.6076E-04	3.4025E-05	-1.4570E-06	-1.3743E-06

In the imaging optical lens system according to the 3rd embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment. Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	4.38	f/R8	0.14
Fno	2.20	f/f2	0.25
HFOV [deg.]	38.2	f/f4	-0.39
V2/V4	1.00	f4/f5	-0.02
CT3/T23	0.61	f/f3 + f/f5	0.04
CT5/CT4	2.11	TD/(2*SD52)	0.71
(T12 + T45)/(T23 + T34)	0.13	TL/lmgH	1.40
R10/R6	0.09		

4th Embodiment

FIG. 4A is a schematic view of an imaging device according to the 4th embodiment of the present disclosure. FIG. 4B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 4th embodiment.

In FIG. 4A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present

disclosure and an image sensor 480. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, an IR-cut filter 460 and an image plane 470, wherein the imaging optical lens system has a total of five lens elements (410-450) with refractive power.

The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 411 and the image-side surface 412 of the first lens element 410 are aspheric.

The second lens element 420 with negative refractive power has an object-side surface 421 being concave in a paraxial region thereof and an image-side surface 422 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 421 and the image-side surface 422 of the second lens element 420 are aspheric.

The third lens element 430 with positive refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 431 and the image-side surface 432 of the third lens element 430 are aspheric.

The fourth lens element 440 with negative refractive power has an object-side surface 441 being concave in a paraxial

17

region thereof and an image-side surface **442** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **441** and the image-side surface **442** of the fourth lens element **440** are aspheric. Furthermore, both of the object-side surface **441** and the image-side surface **442** of the fourth lens element **440** have at least one inflection point.

The fifth lens element **450** with positive refractive power has an object-side surface **451** being convex in a paraxial region thereof and an image-side surface **452** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **451** and the image-side surface **452** of the fifth lens element **450** are aspheric. Furthermore, both of the object-side surface **451** and the image-side surface **452** of

18

the fifth lens element **450** have at least one inflection point, wherein the object-side surface **451** of the fifth lens element **450** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **460** is made of glass and located between the fifth lens element **450** and the image plane **470**, and will not affect the focal length of the imaging optical lens system. The image sensor **480** is disposed on the image plane **470** of the imaging optical lens system.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment								
f = 2.33 mm, Fno = 1.90, HFOV = 42.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.150					
2	Lens 1	1.135	ASP 0.392	Plastic	1.565	56.5	2.45	
3		5.516	ASP 0.057					
4	Lens 2	-10.835	ASP 0.180	Plastic	1.650	21.4	-13.04	
5		39.061	ASP 0.218					
6	Lens 3	9.257	ASP 0.211	Plastic	1.650	21.4	43.90	
7		13.582	ASP 0.292					
8	Lens 4	-45.822	ASP 0.326	Plastic	1.650	21.4	-15.24	
9		12.670	ASP 0.070					
10	Lens 5	1.026	ASP 0.557	Plastic	1.565	56.0	15.64	
11		0.933	ASP 0.300					
12	IR-cut filter	Plano	0.145	Glass	1.517	64.2	—	
13		Plano	0.263					
14	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-1.9052E-01	-1.0000E+00	4.0500E+00	-1.9120E+01	-3.0000E+01
A4 =	-7.8312E-02	-2.3970E-01	-2.1685E-01	-6.8367E-02	-4.9648E-01
A6 =	3.8174E-01	-1.2608E+00	1.5719E-01	6.8334E-01	-3.5596E-01
A8 =	-2.5697E+00	7.2832E+00	4.6612E+00	8.9534E-03	-1.9174E+00
A10 =	5.6527E+00	-2.3597E+01	-1.9856E+01	-4.6163E-01	3.4978E+01
A12 =	-7.9286E+00	3.9222E+01	3.9226E+01	-6.4750E+00	-1.6387E+02
A14 =	2.3256E+00	-2.4238E+01	-2.6534E+01	9.4131E+00	3.1777E+02
A16 =					-2.2422E+02
Surface #	7	8	9	10	11
k =	-2.9043E+01	-3.0000E+01	-1.0000E+00	-1.5605E+00	-3.4293E+00
A4 =	-3.6946E-01	8.0338E-01	3.3490E-01	-8.9129E-01	-3.8997E-01
A6 =	-1.0736E-01	-2.9468E+00	-7.8972E-01	9.6397E-01	4.1235E-01
A8 =	-5.1828E+00	5.1320E+00	8.0126E-01	-6.4921E-01	-3.4925E-01
A10 =	3.2629E+01	-6.5488E+00	-6.4266E-01	2.8490E-01	1.9235E-01
A12 =	-8.5187E+01	4.7810E+00	3.8500E-01	-7.7689E-02	-6.2592E-02
A14 =	1.0437E+02	-1.9300E+00	-1.3250E-01	1.1893E-02	1.0797E-02
A16 =	-4.4753E+01	4.8529E-01	1.8445E-02	-7.7957E-04	-7.5535E-04

19

In the imaging optical lens system according to the 4th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment. Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	2.33	f/R8	0.18
Fno	1.90	f/f2	0.18
HFOV [deg.]	42.9	f/f4	-0.15
V2/V4	1.00	f4/f5	-0.97
CT3/T23	0.97	f/f3 + f/f5	0.20
CT5/CT4	1.71	TD/(2*SD52)	0.62
(T12 + T45)/(T23 + T34)	0.25	TL/lmgH	1.37
R10/R6	0.07		

5th Embodiment

FIG. 5A is a schematic view of an imaging device according to the 5th embodiment of the present disclosure. FIG. 5B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 5th embodiment.

In FIG. 5A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor 580. The imaging optical lens system includes, in order from an object side to an image side, a first lens element 510, an aperture stop 500, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, an IR-cut filter 560 and an image plane 570, wherein the imaging optical lens system has a total of five lens elements (510-550) with refractive power.

The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being convex in a paraxial region thereof, and is made of plastic material. The object-side surface 511 and the image-side surface 512 of the first lens element 510 are aspheric.

20

The second lens element 520 with negative refractive power has an object-side surface 521 being concave in a paraxial region thereof and an image-side surface 522 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 521 and the image-side surface 522 of the second lens element 520 are aspheric.

The third lens element 530 with positive refractive power has an object-side surface 531 being convex in a paraxial region thereof and an image-side surface 532 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 531 and the image-side surface 532 of the third lens element 530 are aspheric.

The fourth lens element 540 with negative refractive power has an object-side surface 541 being concave in a paraxial region thereof and an image-side surface 542 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 541 and the image-side surface 542 of the fourth lens element 540 are aspheric. Furthermore, both of the object-side surface 541 and the image-side surface 542 of the fourth lens element 540 have at least one inflection point.

The fifth lens element 550 with positive refractive power has an object-side surface 551 being convex in a paraxial region thereof and an image-side surface 552 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 551 and the image-side surface 552 of the fifth lens element 550 are aspheric. Furthermore, both of the object-side surface 551 and the image-side surface 552 of the fifth lens element 550 have at least one inflection point, wherein the object-side surface 551 of the fifth lens element 550 is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter 560 is made of glass and located between the fifth lens element 550 and the image plane 570, and will not affect the focal length of the imaging optical lens system. The image sensor 580 is disposed on the image plane 570 of the imaging optical lens system.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 2.59 mm, Fno = 2.60, HFOV = 40.0 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Lens 1	1.144	ASP	0.356	Plastic	1.535	55.7	2.04
2		-20.185	ASP	0.010				
3	Ape. Stop	Plano		0.030				
4	Lens 2	-11.373	ASP	0.182	Plastic	1.650	21.4	-5.67
5		5.482	ASP	0.304				
6	Lens 3	9.494	ASP	0.229	Plastic	1.632	23.4	48.85
7		13.582	ASP	0.348				
8	Lens 4	-11.955	ASP	0.313	Plastic	1.650	21.4	-9.42
9		12.670	ASP	0.070				
10	Lens 5	1.136	ASP	0.626	Plastic	1.535	55.7	48.28
11		0.961	ASP	0.300				
12	IR-cut filter	Plano		0.145	Glass	1.517	64.2	—
13		Plano		0.226				
14	Image	Plano		—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients					
Surface #	1	2	4	5	6
k =	-1.7840E-01	-1.0000E+00	-2.6095E+01	-2.9962E+01	3.4972E+00
A4 =	-6.5246E-02	-1.7686E-01	-1.5792E-01	-1.1491E-01	-4.6839E-01
A6 =	2.3462E-01	-1.2539E-01	6.6867E-01	4.0119E-01	-5.1654E-01
A8 =	-2.3789E+00	6.0328E+00	3.6742E+00	-1.0500E+00	-2.7106E+00
A10 =	6.5919E+00	-3.0979E+01	-3.0436E+01	-2.4418E+00	3.4931E+01
A12 =	-9.5708E+00	5.4433E+01	6.5309E+01	5.5013E-03	-1.6471E+02
A14 =	2.3255E+00	-2.4238E+01	-2.6534E+01	9.4131E+00	3.1777E+02
A16 =					-2.2422E+02
Surface #	7	8	9	10	11
k =	-2.9633E+01	3.3974E+00	-1.0000E+00	-1.4665E+00	-3.3207E+00
A4 =	-3.2948E-01	7.5347E-01	2.9253E-01	-8.9979E-01	-3.9570E-01
A6 =	-9.7686E-02	-2.8708E+00	-7.8990E-01	9.6687E-01	4.1633E-01
A8 =	-5.3370E+00	5.0689E+00	8.1057E-01	-6.4906E-01	-3.5025E-01
A10 =	3.2349E+01	-6.6029E+00	-6.4099E-01	2.8481E-01	1.9217E-01
A12 =	-8.5117E+01	4.7740E+00	3.8498E-01	-7.7726E-02	-6.2639E-02
A14 =	1.0488E+02	-1.8942E+00	-1.3263E-01	1.1889E-02	1.0799E-02
A16 =	-4.4499E+01	5.3768E-01	1.8376E-02	-7.7820E-04	-7.5355E-04

In the imaging optical lens system according to the 5th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment. Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment			
f [mm]	2.59	f/R8	0.20
Fno	2.60	f/f2	0.46
HFOV [deg.]	40.0	f/f4	-0.27
V2/V4	1.00	f4/f5	-0.20
CT3/T23	0.75	f/f3 + f/f5	0.11
CT5/CT4	2.00	TD/(2*SD52)	0.67
(T12 + T45)/(T23 + T34)	0.17	TL/lmgH	1.43
R10/R6	0.07		

6th Embodiment

FIG. 6A is a schematic view of an imaging device according to the 6th embodiment of the present disclosure. FIG. 6B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 6th embodiment.

In FIG. 6A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor 680. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, an IR-cut filter 660, and an image plane 670, wherein the imaging optical lens system has a total of five lens elements (610-650) with refractive power.

The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 611 and the image-side surface 612 of the first lens element 610 are aspheric.

The second lens element 620 with negative refractive power has an object-side surface 621 being concave in a paraxial region thereof and an image-side surface 622 being convex in a paraxial region thereof, and is made of plastic material. The object-side surface 621 and the image-side surface 622 of the second lens element 620 are aspheric.

The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 631 and the image-side surface 632 of the third lens element 630 are aspheric.

The fourth lens element 640 with negative refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 641 and the image-side surface 642 of the fourth lens element 640 are aspheric. Furthermore, both of the object-side surface 641 and the image-side surface 642 of the fourth lens element 640 have at least one inflection point.

The fifth lens element 650 with negative refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 651 and the image-side surface 652 of the fifth lens element 650 are aspheric. Furthermore, both of the object-side surface 651 and the image-side surface 652 of the fifth lens element 650 have at least one inflection point, wherein the object-side surface 651 of the fifth lens element 650 is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter 660 is made of glass and located in order between the fifth lens element 650 and the image plane 670, and will not affect the focal length of the imaging optical lens system. The image sensor 680 is disposed on the image plane 670 of the imaging optical lens system.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment								
f = 3.76 mm, Fno = 2.10, HFOV = 37.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.297					
2	Lens 1	1.332	ASP	0.562	Plastic	1.544	55.9	2.81
3		8.792	ASP	0.140				
4	Lens 2	-3.038	ASP	0.200	Plastic	1.632	23.4	-7.63
5		-8.429	ASP	0.337				
6	Lens 3	7.400	ASP	0.298	Plastic	1.640	23.3	46.01
7		9.729	ASP	0.560				
8	Lens 4	-12.056	ASP	0.355	Plastic	1.632	23.4	-13.69
9		31.023	ASP	0.100				
10	Lens 5	2.009	ASP	0.917	Plastic	1.544	55.9	-27.62
11		1.487	ASP	0.400				
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.274					
14	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-1.1418E-03	-1.0000E+00	-1.4956E+01	-2.3704E+01	-3.0000E+01
A4 =	-3.0285E-02	-4.3728E-02	-4.6482E-02	3.4873E-02	-2.1418E-01
A6 =	1.2421E-01	-3.4680E-01	8.3154E-02	2.5819E-01	-2.2812E-01
A8 =	-3.7320E-01	1.1496E+00	6.9764E-01	6.6608E-03	-1.0228E-01
A10 =	5.1006E-01	-2.1985E+00	-1.8239E+00	5.1780E-02	3.1619E+00
A12 =	-4.1335E-01	2.0141E+00	2.0208E+00	-3.7951E-01	-8.9927E+00
A14 =	4.1498E-02	-7.1908E-01	-7.5267E-01	5.3072E-01	1.0256E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	-3.0000E+01	9.3298E-01	-1.0000E+00	-1.6714E+00	-4.1055E+00
A4 =	-1.6868E-01	3.6295E-01	1.4248E-01	-3.8408E-01	-1.6930E-01
A6 =	-9.5857E-03	-7.3926E-01	-2.0075E-01	2.5401E-01	1.0709E-01
A8 =	-8.4282E-01	7.8023E-01	1.1847E-01	-1.0117E-01	-5.4100E-02
A10 =	2.9916E+00	-5.9882E-01	-5.8052E-02	2.6080E-02	1.7629E-02
A12 =	-4.5466E+00	2.6367E-01	2.0900E-02	-4.1747E-03	-3.3697E-03
A14 =	3.3032E+00	-6.1544E-02	-4.1911E-03	3.7685E-04	3.4166E-04
A16 =	-9.0443E-01	6.9680E-03	3.3448E-04	-1.4739E-05	-1.4087E-05

In the imaging optical lens system according to the 6th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment. Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	3.76	f/R8	0.12
Fno	2.10	f/f2	0.49
HFOV [deg.]	37.9	f/f4	-0.27
V2/V4	1.00	f4/f5	0.50
CT3/T23	0.88	f/f3 + f/f5	0.22
CT5/CT4	2.58	TD/(2*SD52)	0.70
(T12 + T45)/(T23 + T34)	0.27	TL/lmgH	1.45
R10/R6	0.15		

7th Embodiment

FIG. 7A is a schematic view of an imaging device according to the 7th embodiment of the present disclosure. FIG. 7B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 7th embodiment.

In FIG. 7A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor 780. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, an IR-cut filter 760 and an image plane 770, wherein the imaging optical lens system has a total of five lens elements (710-750) with refractive power.

The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof, and is made of plastic material. The object-side surface 711 and the image-side surface 712 of the first lens element 710 are aspheric.

25

The second lens element **720** with negative refractive power has an object-side surface **721** being concave in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **721** and the image-side surface **722** of the second lens element **720** are aspheric.

The third lens element **730** with positive refractive power has an object-side surface **731** being convex in a paraxial region thereof and an image-side surface **732** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **731** and the image-side surface **732** of the third lens element **730** are aspheric.

The fourth lens element **740** with negative refractive power has an object-side surface **741** being concave in a paraxial region thereof and an image-side surface **742** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **741** and the image-side surface **742** of the fourth lens element **740** are aspheric. Furthermore, both of the object-side surface **741** and the image-side surface **742** of the fourth lens element **740** have at least one inflection point.

26

The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex in a paraxial region thereof and an image-side surface **752** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **751** and the image-side surface **752** of the fifth lens element **750** are aspheric. Furthermore, both of the object-side surface **751** and the image-side surface **752** of the fifth lens element **750** have at least one inflection point, wherein the object-side surface **751** of the fifth lens element **750** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **760** is made of glass and located in order between the fifth lens element **750** and the image plane **770**, and will not affect the focal length of the imaging optical lens system. The image sensor **780** is disposed on the image plane **770** of the imaging optical lens system.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment								
f = 3.67 mm, Fno = 2.10, HFOV = 38.7 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.270					
2	Lens 1	1.390	ASP	0.504	Plastic	1.544	55.9	2.75
3		16.807	ASP	0.119				
4	Lens 2	-7.641	ASP	0.211	Plastic	1.632	23.4	-4.87
5		5.208	ASP	0.298				
6	Lens 3	3.494	ASP	0.298	Plastic	1.607	26.6	11.70
7		6.654	ASP	0.595				
8	Lens 4	-5.828	ASP	0.341	Plastic	1.632	23.4	-6.66
9		15.523	ASP	0.050				
10	Lens 5	1.509	ASP	0.995	Plastic	1.544	55.9	11.78
11		1.515	ASP	0.400				
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2		—
13		Plano	0.333					
14	Image	Plano	—					

Note:
Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	1.1628E-02	-1.0000E+00	-1.3246E+01	-3.0000E+01	-5.5124E-01
A4 =	-2.8889E-02	-3.0238E-02	-4.9490E-02	-2.3407E-02	-1.9597E-01
A6 =	1.1835E-01	-2.9631E-01	8.3107E-02	2.4462E-01	-1.9694E-01
A8 =	-3.8046E-01	1.1598E+00	7.0936E-01	-1.2686E-02	-8.8463E-02
A10 =	4.8521E-01	-2.2118E+00	-1.8373E+00	4.8880E-02	3.1190E+00
A12 =	-3.8541E-01	1.9851E+00	2.0226E+00	-4.1231E-01	-9.0502E+00
A14 =	7.8857E-02	-6.8082E-01	-7.6155E-01	4.9115E-01	1.0300E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	5.0000E+00	-2.9568E+01	-1.0000E+00	-1.3349E+00	-2.4765E+00
A4 =	-1.2376E-01	4.0402E-01	1.3164E-01	-3.9047E-01	-1.8196E-01
A6 =	-1.4439E-02	-7.6778E-01	-1.8651E-01	2.5319E-01	1.0912E-01
A8 =	-8.5592E-01	7.9119E-01	1.1563E-01	-1.0116E-01	-5.4256E-02
A10 =	2.9880E+00	-5.9476E-01	-5.8206E-02	2.6094E-02	1.7630E-02
A12 =	-4.5461E+00	2.6342E-01	2.0940E-02	-4.1728E-03	-3.3647E-03
A14 =	3.3046E+00	-6.1983E-02	-4.1852E-03	3.7669E-04	3.4089E-04
A16 =	-9.0365E-01	6.6830E-03	3.3242E-04	-1.4734E-05	-1.4052E-05

27

In the imaging optical lens system according to the 7th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment. Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment			
f [mm]	3.67	f/R8	0.24
Fno	2.10	f/f2	0.75
HFOV [deg.]	38.7	f/f4	-0.55
V2/V4	1.00	f4/f5	-0.57
CT3/T23	1.00	f/f3 + f/f5	0.62
CT5/CT4	2.92	TD/(2*SD52)	0.69
(T12 + T45)/(T23 + T34)	0.19	TL/lmgH	1.45
R10/R6	0.23		

8th Embodiment

FIG. 8A is a schematic view of an imaging device according to the 8th embodiment of the present disclosure. FIG. 8B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 8th embodiment.

In FIG. 8A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor **880**. The imaging optical lens system includes, in order from an object side to an image side, a first lens element **810**, an aperture stop **800**, a second lens element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, an IR-cut filter **860** and an image plane **870**, wherein the imaging optical lens system has a total of five lens elements (**810-850**) with refractive power.

The first lens element **810** with positive refractive power has an object-side surface **811** being convex in a paraxial region thereof and an image-side surface **812** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **811** and the image-side surface **812** of the first lens element **810** are aspheric.

28

The second lens element **820** with negative refractive power has an object-side surface **821** being concave in a paraxial region thereof and an image-side surface **822** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **821** and the image-side surface **822** of the second lens element **820** are aspheric.

The third lens element **830** with positive refractive power has an object-side surface **831** being convex in a paraxial region thereof and an image-side surface **832** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **831** and the image-side surface **832** of the third lens element **830** are aspheric.

The fourth lens element **840** with negative refractive power has an object-side surface **841** being concave in a paraxial region thereof and an image-side surface **842** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **841** and the image-side surface **842** of the fourth lens element **840** are aspheric. Furthermore, both of the object-side surface **841** and the image-side surface **842** of the fourth lens element **840** have at least one inflection point.

The fifth lens element **850** with positive refractive power has an object-side surface **851** being convex in a paraxial region thereof and an image-side surface **852** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **851** and the image-side surface **852** of the fifth lens element **850** are aspheric. Furthermore, both of the object-side surface **851** and the image-side surface **852** of the fifth lens element **850** have at least one inflection point, wherein the object-side surface **851** of the fifth lens element **850** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **860** is made of glass and located between the fifth lens element **850** and the image plane **870**, and will not affect the focal length of the imaging optical lens system. The image sensor **880** is disposed on the image plane **870** of the imaging optical lens system.

The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment								
f = 3.70 mm, Fno = 2.40, HFOV = 38.5 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Lens 1	1.391	ASP	0.521	Plastic	1.544	55.9	2.71
2		20.933	ASP	0.032				
3	Ape. Stop	Plano		0.037				
4	Lens 2	-16.187	ASP	0.202	Plastic	1.639	23.5	-5.73
5		4.751	ASP	0.431				
6	Lens 3	5.313	ASP	0.267	Plastic	1.607	26.6	28.28
7		7.547	ASP	0.553				
8	Lens 4	-10.560	ASP	0.400	Plastic	1.632	23.4	-9.82
9		15.261	ASP	0.050				
10	Lens 5	1.611	ASP	0.956	Plastic	1.544	55.9	28.35
11		1.423	ASP	0.400				
12	IR-cut filter	Plano		0.210	Glass	1.517	64.2	—
13		Plano		0.296				
14	Image	Plano		—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients					
Surface #	1	2	4	5	6
k =	5.9134E-02	-1.0000E+00	-1.7363E+01	5.0000E+00	4.1156E+00
A4 =	-2.3601E-02	-5.8920E-02	-5.5000E-02	5.4103E-03	-1.6703E-01
A6 =	1.0175E-01	-2.3133E-01	1.0160E-01	2.1092E-01	-2.2713E-01
A8 =	-3.8078E-01	1.1755E+00	7.1023E-01	-2.8395E-02	-9.9135E-02
A10 =	5.2200E-01	-2.2818E+00	-1.8645E+00	3.2726E-02	3.1783E+00
A12 =	-3.4662E-01	1.9300E+00	1.8799E+00	-4.3682E-01	-9.0221E+00
A14 =	-3.9034E-03	-6.5437E-01	-6.8877E-01	5.0594E-01	1.0195E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	2.8907E+00	5.0000E+00	-1.0000E+00	-1.4582E+00	-3.1690E+00
A4 =	-1.4054E-01	3.9170E-01	1.3410E-01	-3.9409E-01	-1.7822E-01
A6 =	-1.4953E-02	-7.5113E-01	-1.8473E-01	2.5338E-01	1.1037E-01
A8 =	-8.4188E-01	7.9092E-01	1.1464E-01	-1.0107E-01	-5.4601E-02
A10 =	2.9826E+00	-5.9565E-01	-5.8350E-02	2.6105E-02	1.7621E-02
A12 =	-4.5562E+00	2.6294E-01	2.0971E-02	-4.1735E-03	-3.3660E-03
A14 =	3.2984E+00	-6.2139E-02	-4.1776E-03	3.7651E-04	3.4152E-04
A16 =	-9.0163E-01	6.6570E-03	3.3625E-04	-1.4766E-05	-1.4059E-05

In the imaging optical lens system according to the 8th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment. Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	3.70	f/R8	0.24
Fno	2.40	f/f2	0.65
HFOV [deg.]	38.5	f/f4	-0.38
V2/V4	1.00	f4/f5	-0.35
CT3/T23	0.62	f/f3 + f/f5	0.26
CT5/CT4	2.39	TD/(2*SD52)	0.71
(T12 + T45)/(T23 + T34)	0.12	TL/lmgH	1.45
R10/R6	0.19		

9th Embodiment

FIG. 9A is a schematic view of an imaging device according to the 9th embodiment of the present disclosure. FIG. 9B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 9th embodiment.

In FIG. 9A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor **980**. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop **900**, a first lens element **910**, a second lens element **920**, a third lens element **930**, a fourth lens element **940**, a fifth lens element **950**, an IR-cut filter **960** and an image plane **970**, wherein the imaging optical lens system has a total of five lens elements (**910-950**) with refractive power.

The first lens element **910** with positive refractive power has an object-side surface **911** being convex in a paraxial region thereof and an image-side surface **912** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **911** and the image-side surface **912** of the first lens element **910** are aspheric.

The second lens element **920** with negative refractive power has an object-side surface **921** being convex in a paraxial region thereof and an image-side surface **922** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **921** and the image-side surface **922** of the second lens element **920** are aspheric.

The third lens element **930** with positive refractive power has an object-side surface **931** being convex in a paraxial region thereof and an image-side surface **932** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **931** and the image-side surface **932** of the third lens element **930** are aspheric.

The fourth lens element **940** with positive refractive power has an object-side surface **941** being convex in a paraxial region thereof and an image-side surface **942** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **941** and the image-side surface **942** of the fourth lens element **940** are aspheric. Furthermore, both of the object-side surface **941** and the image-side surface **942** of the fourth lens element **940** have at least one inflection point.

The fifth lens element **950** with negative refractive power has an object-side surface **951** being convex in a paraxial region thereof and an image-side surface **952** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **951** and the image-side surface **952** of the fifth lens element **950** are aspheric. Furthermore, both of the object-side surface **951** and the image-side surface **952** of the fifth lens element **950** have at least one inflection point, wherein the object-side surface **951** of the fifth lens element **950** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **960** is made of glass and located between the fifth lens element **950** and the image plane **970**, and will not affect the focal length of the imaging optical lens system. The image sensor **980** is disposed on the image plane **970** of the imaging optical lens system.

The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

TABLE 17

9th Embodiment							
f = 3.59 mm, Fno = 2.20, HFOV = 38.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.283				
2	Lens 1	1.309	ASP	0.499	Plastic	1.543	56.5
3		13.624	ASP	0.057			
4	Lens 2	11.877	ASP	0.200	Plastic	1.639	23.5
5		2.458	ASP	0.439			
6	Lens 3	17.666	ASP	0.321	Plastic	1.543	56.5
7		66.553	ASP	0.463			
8	Lens 4	9.220	ASP	0.482	Plastic	1.543	56.5
9		31.023	ASP	0.130			
10	Lens 5	1.829	ASP	0.780	Plastic	1.535	55.7
11		1.159	ASP	0.400			
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
13		Plano	0.266				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 18

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	1.0326E-01	-1.0000E+00	-2.9871E+01	-5.7016E+00	-2.9475E+01
A4 =	-1.2596E-02	-3.8565E-02	-1.0704E-01	1.8888E-02	-1.7826E-01
A6 =	1.3045E-01	-1.6302E-01	1.2686E-01	2.2849E-01	-1.7585E-01
A8 =	-4.0347E-01	1.1726E+00	7.3452E-01	2.4296E-03	-1.8990E-01
A10 =	5.7896E-01	-2.4398E+00	-2.0235E+00	-3.1513E-02	3.2006E+00
A12 =	-2.4633E-01	2.0984E+00	1.7430E+00	-4.4074E-01	-8.8834E+00
A14 =	-1.1079E-01	-7.6606E-01	-4.7771E-01	7.1968E-01	1.0135E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	-1.0000E+00	-1.0000E+00	-1.0000E+00	-1.1533E+00	-4.4655E+00
A4 =	-1.5665E-01	3.1818E-01	1.5550E-01	-4.0010E-01	-1.6513E-01
A6 =	-3.1134E-02	-6.8550E-01	-1.8486E-01	2.5334E-01	1.0858E-01
A8 =	-8.1771E-01	7.7158E-01	1.1344E-01	-1.0087E-01	-5.4899E-02
A10 =	2.9803E+00	-5.9404E-01	-5.9042E-02	2.6127E-02	1.7648E-02
A12 =	-4.5556E+00	2.6623E-01	2.1119E-02	-4.1752E-03	-3.3489E-03
A14 =	3.3077E+00	-6.0931E-02	-4.1353E-03	3.7449E-04	3.4147E-04
A16 =	-9.0858E-01	5.5365E-03	3.3115E-04	-1.4544E-05	-1.4322E-05

In the imaging optical lens system according to the 9th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment. Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

9th Embodiment			
f [mm]	3.59	f/R8	0.12
Fno	2.20	f/f2	0.73
HFOV [deg.]	38.1	f/f4	0.15
V2/V4	0.42	f4/f5	-2.40
CT3/T23	0.73	f/f3 + f/f5	0.44
CT5/CT4	1.62	TD/(2*SD52)	0.71
(T12 + T45)/(T23 + T34)	0.21	TL/lmgH	1.49
R10/R6	0.02		

10th Embodiment

FIG. 10A is a schematic view of an imaging device according to the 10th embodiment of the present disclosure. FIG. 10B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 10th embodiment.

In FIG. 10A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor 1080. The imaging optical lens system includes, in order from an object side to an image side, an aperture stop 1000, a first lens element 1010, a second lens element 1020, a third lens element 1030, a fourth lens element 1040, a fifth lens element 1050, an IR-cut filter 1060 and an image plane 1070, wherein the imaging optical lens system has a total of five lens elements (1010-1050) with refractive power.

The first lens element 1010 with positive refractive power has an object-side surface 1011 being convex in a paraxial region thereof and an image-side surface 1012 being concave in a paraxial region thereof, and is made of plastic material.

33

The object-side surface **1011** and the image-side surface **1012** of the first lens element **1010** are aspheric.

The second lens element **1020** with negative refractive power has an object-side surface **1021** being convex in a paraxial region thereof and an image-side surface **1022** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1021** and the image-side surface **1022** of the second lens element **1020** are aspheric.

The third lens element **1030** with positive refractive power has an object-side surface **1031** being convex in a paraxial region thereof and an image-side surface **1032** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1031** and the image-side surface **1032** of the third lens element **1030** are aspheric.

The fourth lens element **1040** with negative refractive power has an object-side surface **1041** being convex in a paraxial region thereof and an image-side surface **1042** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1041** and the image-side surface **1042** of the fourth lens element **1040** are aspheric. Furthermore, both of the object-side surface **1041** and the image-side surface **1042** of the fourth lens element **1040** have at least one inflection point.

34

The fifth lens element **1050** with negative refractive power has an object-side surface **1051** being convex in a paraxial region thereof and an image-side surface **1052** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1051** and the image-side surface **1052** of the fifth lens element **1050** are aspheric. Furthermore, both of the object-side surface **1051** and the image-side surface **1052** of the fifth lens element **1050** have at least one inflection point, wherein the object-side surface **1051** of the fifth lens element **1050** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **1060** is made of glass and located between the fifth lens element **1050** and the image plane **1070**, and will not affect the focal length of the imaging optical lens system. The image sensor **1080** is disposed on the image plane **1070** of the imaging optical lens system.

The detailed optical data of the 10th embodiment are shown in Table 19 and the aspheric surface data are shown in Table 20 below.

TABLE 19

10th Embodiment							
f = 3.63 mm, Fno = 2.25, HFOV = 37.7 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.278				
2	Lens 1	1.314 ASP	0.514	Plastic	1.544	55.9	2.53
3		25.705 ASP	0.048				
4	Lens 2	15.628 ASP	0.200	Plastic	1.639	23.5	-4.64
5		2.480 ASP	0.446				
6	Lens 3	13.499 ASP	0.314	Plastic	1.530	55.8	44.73
7		31.104 ASP	0.448				
8	Lens 4	6.252 ASP	0.405	Plastic	1.530	55.8	-27.30
9		4.268 ASP	0.122				
10	Lens 5	1.865 ASP	0.956	Plastic	1.544	55.9	-34.81
11		1.391 ASP	0.400				
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
13		Plano	0.205				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 20

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	1.1642E-01	-1.0000E+00	-2.9797E+01	-6.2612E+00	-1.8431E+01
A4 =	-1.4044E-02	-3.5337E-02	-1.0558E-01	1.4503E-02	-1.7472E-01
A6 =	1.2828E-01	-1.4801E-01	1.1951E-01	2.3218E-01	-1.7634E-01
A8 =	-4.0184E-01	1.1241E+00	7.7335E-01	-2.6046E-02	-1.8589E-01
A10 =	5.6647E-01	-2.3674E+00	-2.0966E+00	-5.7589E-02	3.1979E+00
A12 =	-2.4428E-01	1.9963E+00	1.7850E+00	-2.7838E-01	-8.8977E+00
A14 =	-1.0845E-01	-6.6675E-01	-4.1662E-01	6.0278E-01	1.0139E+01
A16 =					-4.1666E+00
Surface #	7	8	9	10	11
k =	-1.0000E+00	-1.0000E+00	-1.0000E+00	-8.9508E-01	-4.1394E+00
A4 =	-1.5774E-01	2.9947E-01	8.6214E-02	-3.9527E-01	-1.6898E-01
A6 =	-2.3956E-02	-6.9723E-01	-1.6948E-01	2.5256E-01	1.0990E-01
A8 =	-8.2036E-01	7.7180E-01	1.1392E-01	-1.0098E-01	-5.4908E-02
A10 =	2.9812E+00	-5.9257E-01	-5.9197E-02	2.6127E-02	1.7620E-02
A12 =	-4.5519E+00	2.6603E-01	2.1072E-02	-4.1766E-03	-3.3524E-03
A14 =	3.3065E+00	-6.0995E-02	-4.1404E-03	3.7493E-04	3.4178E-04
A16 =	-9.0885E-01	5.5910E-03	3.3067E-04	-1.4541E-05	-1.4256E-05

35

In the imaging optical lens system according to the 10th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment. Moreover, these parameters can be calculated from Table 19 and Table 20 as the following values and satisfy the following conditions:

10th Embodiment			
f [mm]	3.63	f/R8	0.85
Fno	2.25	f/f2	0.78
HFOV [deg.]	37.7	f/f4	-0.13
V2/V4	0.42	f4/f5	0.78
CT3/T23	0.70	f/f3 + f/f5	0.19
CT5/CT4	2.36	TD/(2*SD52)	0.73
(T12 + T45)/(T23 + T34)	0.19	TL/lmgH	1.49
R10/R6	0.04		

11th Embodiment

FIG. 11A is a schematic view of an imaging device according to the 11th embodiment of the present disclosure. FIG. 11B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 11th embodiment.

In FIG. 11A, the imaging device includes the imaging optical lens system (not otherwise herein labeled) of the present disclosure and an image sensor **1180**. The imaging optical lens system includes, in order from an object side to an image side, a first lens element **1110**, an aperture stop **1100**, a second lens element **1120**, a third lens element **1130**, a fourth lens element **1140**, a fifth lens element **1150**, an IR-cut filter **1160** and an image plane **1170**, wherein the imaging optical lens system has a total of five lens elements (**1110-1150**) with refractive power.

The first lens element **1110** with positive refractive power has an object-side surface **1111** being convex in a paraxial region thereof and an image-side surface **1112** being convex in a paraxial region thereof, and is made of glass material. The object-side surface **1111** and the image-side surface **1112** of the first lens element **1110** are aspheric.

36

The second lens element **1120** with negative refractive power has an object-side surface **1121** being concave in a paraxial region thereof and an image-side surface **1122** being convex in a paraxial region thereof, and is made of plastic material. The object-side surface **1121** and the image-side surface **1122** of the second lens element **1120** are aspheric.

The third lens element **1130** with positive refractive power has an object-side surface **1131** being convex in a paraxial region thereof and an image-side surface **1132** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1131** and the image-side surface **1132** of the third lens element **1130** are aspheric.

The fourth lens element **1140** with positive refractive power has an object-side surface **1141** being convex in a paraxial region thereof and an image-side surface **1142** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1141** and the image-side surface **1142** of the fourth lens element **1140** are aspheric. Furthermore, both of the object-side surface **1141** and the image-side surface **1142** of the fourth lens element **1140** have at least one inflection point.

The fifth lens element **1150** with negative refractive power has an object-side surface **1151** being convex in a paraxial region thereof and an image-side surface **1152** being concave in a paraxial region thereof, and is made of plastic material. The object-side surface **1151** and the image-side surface **1152** of the fifth lens element **1150** are aspheric. Furthermore, both of the object-side surface **1151** and the image-side surface **1152** of the fifth lens element **1150** have at least one inflection point, wherein the object-side surface **1151** of the fifth lens element **1150** is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

The IR-cut filter **1160** is made of glass and located between the fifth lens element **1150** and the image plane **1170**, and will not affect the focal length of the imaging optical lens system. The image sensor **1080** is disposed on the image plane **1170** of the imaging optical lens system.

The detailed optical data of the 11th embodiment are shown in Table 21 and the aspheric surface data are shown in Table 22 below.

TABLE 21

11th Embodiment							
f = 3.76 mm, Fno = 2.40, HFOV = 37.2 deg.							
Surface #	Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.782	ASP	0.489	Glass	1.548	45.7
2		-5.687	ASP	-0.076			2.53
3	Ape. Stop	Plano	0.146				
4	Lens 2	-2.254	ASP	0.265	Plastic	1.650	21.4
5		-17.240	ASP	0.145			-4.02
6	Lens 3	1.479	ASP	0.272	Plastic	1.650	21.4
7		1.526	ASP	0.588			22.44
8	Lens 4	7.930	ASP	0.425	Plastic	1.544	55.9
9		13.292	ASP	0.522			35.14
10	Lens 5	1.860	ASP	0.687	Plastic	1.544	55.9
11		1.384	ASP	0.500			-20.21
12	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
13		Plano	0.240				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 22

Aspheric Coefficients					
Surface #	1	2	4	5	6
k =	-2.9715E+00	-1.0000E+00	-1.0416E+01	1.0000E+01	-1.4067E+00
A4 =	-3.7698E-03	-1.2176E-01	1.3986E-01	1.9620E-01	-1.6209E-01
A6 =	-2.5351E-02	-1.4762E-01	-1.2238E-01	-9.4454E-02	1.6884E-01
A8 =	-2.2333E-01	7.6119E-01	4.5795E-01	1.8698E-01	-2.4027E-01
A10 =	1.3933E-01	-1.2957E+00	-4.6177E-01	-3.5494E-01	2.5213E-01
A12 =	-1.8116E-01	8.5911E-01	1.9351E-01	2.1835E-01	-1.2596E-01
A14 =	1.6236E-01	-1.1867E-01			
Surface #	7	8	9	10	11
k =	-2.2337E+00	-1.0000E+00	1.0000E+01	-7.4867E-01	-1.1910E+00
A4 =	-5.4191E-02	-5.4528E-02	-1.4685E-01	-2.5197E-01	-1.9585E-01
A6 =	3.3702E-02	3.5980E-02	1.0683E-01	1.8688E-02	6.5425E-02
A8 =	-1.1557E-02	-5.9411E-02	-4.0944E-02	2.0396E-02	-1.8482E-02
A10 =	1.0388E-02	6.5311E-02	6.9281E-03	-4.0643E-03	3.4962E-03
A12 =	2.0149E-02	-4.7776E-02	1.0756E-05	-2.2322E-04	-3.7876E-04
A14 =	-1.7346E-02	1.2861E-02	-2.5244E-04	6.7106E-05	1.7237E-05

In the imaging optical lens system according to the 11th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 11th embodiment. Moreover, these parameters can be calculated from Table 21 and Table 22 as the following values and satisfy the following conditions:

11th Embodiment			
f [mm]	3.76	f/R8	0.28
Fno	2.40	f/f2	0.94
HFOV [deg.]	37.2	f/f4	0.11
V2/V4	0.38	f4/f5	-1.74
CT3/T23	1.88	f/f3 + f/f5	0.35
CT5/CT4	1.62	TD/(2*SD52)	0.75
(T12 + T45)/(T23 + T34)	0.81	TL/lmgH	1.53
R10/R6	0.91		

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-22 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An imaging optical lens system comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element having refractive power;

a third lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof;

a fourth lens element with refractive power having an image-side surface being concave in a paraxial region thereof; and

a fifth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein both of an object-side surface and the image-side surface of the fifth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point;

wherein the imaging optical lens system has a total of five lens elements with refractive power, a focal length of the imaging optical lens system is f, a curvature radius of the image-side surface of the fourth lens element is R8, a central thickness of the fifth lens element is CT5, a central thickness of the fourth lens element is CT4, a focal length of the fourth lens element is f4, an axial distance between the second lens element and the third lens element is T23, a central thickness of the third lens element is CT3, and the following conditions are satisfied:

$$0 < f/R8 < 2.0;$$

$$1.3 < CT5/CT4 < 3.5;$$

$$-0.7 < f/f4 \leq 0.15; \text{ and}$$

$$0.2 < CT3/T23 \leq 1.00.$$

2. The imaging optical lens system of claim 1, wherein the second lens element has negative refractive power, an axial distance between the first lens element and the second lens element is T12, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the third lens element and the fourth lens element is T34, and the following condition is satisfied:

$$0 < (T12 + T45)/(T23 + T34) < 0.40.$$

3. The imaging optical lens system of claim 2, wherein the object-side surface of the fifth lens element is convex in a paraxial region thereof.

4. The imaging optical lens system of claim 2, wherein the first lens element has an image-side surface being concave in a paraxial region thereof.

5. The imaging optical lens system of claim 2, wherein the fourth lens element has an object-side surface being concave in a paraxial region thereof.

39

6. The imaging optical lens system of claim 2, wherein an axial distance between the object-side surface of the first lens element and the image-side surface of the fifth lens element is TD, a vertical distance between a maximum effective radius position on the image-side surface of the fifth lens element and an optical axis is SD52, and the following condition is satisfied:

$$0.50 < TD / (2 * SD52) < 0.80.$$

7. The imaging optical lens system of claim 2, wherein an axial distance between the object-side surface of the first lens element and the image plane is TL, a maximum image height of the imaging optical lens system is ImgH, and the following condition is satisfied:

$$TL / \text{ImgH} < 1.7.$$

8. The imaging optical lens system of claim 1, wherein a curvature radius of the image-side surface of the fifth lens element is R10, a curvature radius of the image-side surface of the third lens element is R6, and the following condition is satisfied:

$$0 < R10 / R6 < 1.0.$$

9. The imaging optical lens system of claim 8, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, and the following condition is satisfied:

$$0 < (T12 + T45) / (T23 + T34) < 0.40.$$

10. The imaging optical lens system of claim 1, wherein an Abbe number of the second lens element is V2, an Abbe number of the fourth lens element is V4, and the following condition is satisfied:

$$0.8 < V2 / V4 < 1.3.$$

11. The imaging optical lens system of claim 1, wherein the central thickness of the fifth lens element is CT5, the central thickness of the fourth lens element is CT4, and the following condition is satisfied:

$$1.5 < CT5 / CT4 < 3.0.$$

12. The imaging optical lens system of claim 3, wherein the focal length of the imaging optical lens system is f, a focal length of the third lens element is f3, a focal length of the fifth lens element is f5, and the following condition is satisfied:

$$0 < |f/f3| + |f/f5| < 0.5.$$

13. An imaging device, comprising:
the imaging optical lens system of claim 1; and
an image sensor.

14. A mobile terminal, comprising:
the imaging device of claim 13.

15. An imaging optical lens system comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element having refractive power;

a third lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof;

a fourth lens element with refractive power having an image-side surface being concave in a paraxial region

40

thereof, wherein both of an object-side surface and the image-side surface of the fourth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the fourth lens element has at least one inflection point; and

a fifth lens element with refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein both of the object-side surface and the image-side surface of the fifth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point;

wherein the imaging optical lens system has a total of five lens elements with refractive power, a focal length of the imaging optical lens system is f, a curvature radius of the image-side surface of the fourth lens element is R8, a focal length of the second lens element is f2, a focal length of the fourth lens element is f4, a focal length of the fifth lens element is f5, and the following conditions are satisfied:

$$0.14 \leq f/R8 \leq 0.85;$$

$$|f/f2| < 0.8; \text{ and}$$

$$-3 < f4/f5 < 1.$$

16. The imaging optical lens system of claim 15, wherein the second lens element has negative refractive power.

17. The imaging optical lens system of claim 16, wherein an Abbe number of the second lens element is V2, an Abbe number of the fourth lens element is V4, and the following condition is satisfied:

$$0.8 < V2 / V4 < 1.3.$$

18. The imaging optical lens system of claim 15, wherein the focal length of the imaging optical lens system is f, the focal length of the fourth lens element is f4, and the following condition is satisfied:

$$-0.6 < f/f4 < 0.$$

19. The imaging optical lens system of claim 15, wherein the focal length of the imaging optical lens system is f, a focal length of the third lens element is f3, the focal length of the fifth lens element is f5, and the following condition is satisfied:

$$0 < |f/f3| + |f/f5| < 0.5.$$

20. The imaging optical lens system of claim 15, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, and the following condition is satisfied:

$$0 < (T12 + T45) / (T23 + T34) < 0.40.$$

21. The imaging optical lens system of claim 15, wherein the object-side surface of the fifth lens element is convex in a paraxial region thereof and changes to a concave shape and then a convex shape from the paraxial region to a peripheral region.

22. The imaging optical lens system of claim 15, wherein a curvature radius of the image-side surface of the fifth lens element is R10, a curvature radius of the image-side surface of the third lens element is R6, and the following condition is satisfied:

$$0 < R10 / R6 < 0.5.$$

41

23. An imaging device, comprising:
the imaging optical lens system of claim 15; and
an image sensor.

24. A mobile terminal, comprising:
the imaging device of claim 23.

25. An imaging optical lens system comprising, in order
from an object side to an image side:

a first lens element with positive refractive power having an
object-side surface being convex in a paraxial region
thereof;

a second lens element having refractive power;

a third lens element with positive refractive power having
an object-side surface being convex in a paraxial region
thereof and an image-side surface being concave in a
paraxial region thereof;

a fourth lens element with refractive power having an
object-side surface being convex in a paraxial region
thereof and an image-side surface being concave in a
paraxial region thereof, wherein both of an object-side
surface and the image-side surface of the fourth lens
element are aspheric, and at least one of the object-side
surface and the image-side surface of the fourth lens
element has at least one inflection point; and

a fifth lens element with refractive power having an object-
side surface being convex in a paraxial region thereof
and an image-side surface being concave in a paraxial
region thereof, wherein both of the object-side surface
and the image-side surface of the fifth lens element are
aspheric, and at least one of the object-side surface and
the image-side surface of the fifth lens element has at
least one inflection point;

wherein the imaging optical lens system has a total of five
lens elements with refractive power, a focal length of the
imaging optical lens system is f , a curvature radius of the
image-side surface of the fourth lens element is $R8$, a
focal length of the second lens element is $f2$, a focal
length of the fourth lens element is $f4$, a focal length of
the fifth lens element is $f5$, and the following conditions
are satisfied:

$$0 < f/R8 \leq 0.85;$$

$$|f/f2| < 0.8; \text{ and}$$

$$-3 < f4/f5 < 1.$$

26. The imaging optical lens system of claim 25, wherein
the second lens element has negative refractive power.

42

27. The imaging optical lens system of claim 26, wherein
an Abbe number of the second lens element is $V2$, an Abbe
number of the fourth lens element is $V4$, and the following
condition is satisfied:

$$0.8 < V2/V4 < 1.3.$$

28. The imaging optical lens system of claim 25, wherein
the focal length of the imaging optical lens system is f , the
focal length of the fourth lens element is $f4$, and the following
condition is satisfied:

$$-0.6 < f/f4 < 0.$$

29. The imaging optical lens system of claim 25, wherein
the focal length of the imaging optical lens system is f , a focal
length of the third lens element is $f3$, the focal length of the
fifth lens element is $f5$, and the following condition is satis-
fied:

$$0 < |f/f3| + |f/f5| < 0.5.$$

30. The imaging optical lens system of claim 25, wherein
an axial distance between the first lens element and the second
lens element is $T12$, an axial distance between the fourth lens
element and the fifth lens element is $T45$, an axial distance
between the second lens element and the third lens element is
 $T23$, an axial distance between the third lens element and the
fourth lens element is $T34$, and the following condition is
satisfied:

$$0 < (T12 + T45)/(T23 + T34) < 0.40.$$

31. The imaging optical lens system of claim 25, wherein
the object-side surface of the fifth lens element is convex in a
paraxial region thereof, changes to a concave shape from the
paraxial region to a peripheral region, and a convex shape in
the peripheral region.

32. The imaging optical lens system of claim 25, wherein a
curvature radius of the image-side surface of the fifth lens
element is $R10$, a curvature radius of the image-side surface
of the third lens element is $R6$, and the following condition is
satisfied:

$$0 < R10/R6 < 0.5.$$

33. An imaging device, comprising:
the imaging optical lens system of claim 25; and
an image sensor.

34. A mobile terminal, comprising:
the imaging device of claim 33.

* * * * *